

Perceive, Externalize, Synthesize: Why Two-Pass Decomposition Beats End-to-End Multimodal Review

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Abstract

The default way to put a multimodal model on a long-form review task is to hand it the raw source—the audio, the page images—and let it reason end-to-end. We give counter-intuitive evidence that this is the wrong default. **Decomposing the task into two same-weights passes is systematically more faithful and more complete:** one pass transcribes the source to text, the other writes the review from that transcript alone—even though the second pass sees strictly *less* than the end-to-end pass. Across a 21-cell suite on Gemini 3.1 Pro (11 audio sources, 10 papers, 9–176 pp), decomposition reduces hallucinations on 18/21 cells and improves coverage on 16/21 (both sign-test significant), and the gain scales inversely with the end-to-end baseline ($r = -0.45$): the more the monolithic pass satisfices, the more decomposition recovers. A nine-experiment mechanistic study then asks *why*, and the cause is not what one might assume. It is **not** a perception failure—the model’s own Pass-1 transcript holds 99.7% of the content the end-to-end pass drops—and **not** that text is an intrinsically better reasoning substrate—identical content is read, and reasoned over, equally well from text and from an *image* of that text, both on open VLMs at the activation level (a layer-wise logit-lens) and behaviorally on the headline frontier models themselves, where with reasoning headroom the image is if anything *better* than text. The cause is a **generation-side bottleneck**: one pass cannot simultaneously perceive, reason, and write a long faithful review, so it satisfices and drops roughly a third of what it saw. Decomposition helps because it gives perception and synthesis each a dedicated, full-budget pass; collapsing the two into one call crashes coverage. The account predicts exactly two failure modes, both of which appear: a *compression bottleneck* when synthesis cannot fit the transcript in its output budget (Mode A), and a *training-prior leak* when removing the source lets synthesis confabulate (Mode B), which a quote-grounded iterative-refinement variant suppresses where it binds. We replicate across vendors (Claude Opus 4.7, MIMO v2-omni), two judges, multi-seed re-runs, and probe/reference ablations. Perceive, externalize, then synthesize. The win is decomposition, not the modality.

Code: <https://github.com/19PINE-AI/cascade-gap>
Website: <https://01.me/research/cascade-gap>

1 Introduction

Multimodal foundation models can read audio and document images directly. The intuitive way to deploy them on a long-form review task is therefore end-to-end: feed the model the

raw source and ask for the review in one call. Keeping the original modality in context looks strictly safer: no transcription step to lose information, and the model sees everything. We argue, and measure, that this intuition is backwards for long-form review.

The alternative is a same-weights two-pass decomposition: the model first transcribes the source to text (Pass-1, *perceive and externalize*), then writes the review from that transcript alone (Pass-2, *synthesize*, with the same weights and the raw modality removed from context). This wins. The obvious explanation, that text is simply a better substrate for the model’s reasoning, turns out to be wrong. We run a nine-experiment mechanistic study (§4) that isolates the cause:

- **Not perception.** The model’s own Pass-1 transcript covers 99.9% of probes. Of the 362 probes the end-to-end pass drops, 361 (99.7%) are present in that transcript. The model perceives the content, then loses it downstream.
- **Not substrate.** On an open VLM, a layer-wise logit-lens finds identical propositional content is read equally well from text and from an *image* of that text, whether one fact or sixteen packed facts, with the same answer-token log-prob and the same readout depth. The same null holds *behaviorally on the headline frontier models*: Gemini 3.1 Pro and Claude Opus 4.7 retrieve a queried fact from pixels and from text with identical accuracy up to 48 packed facts (E8), and the equivalence extends from retrieval to *multi-hop reasoning* (E9, summing six packed facts)—where, on the one model with reasoning headroom, the image is reliably *better* than text, the opposite of H1. A degraded-image control that the same probes flag as non-equivalent confirms the test could detect a gap if one existed. There is no intrinsic “pixels are harder to reason over” penalty for legible content.
- **It is generation under load.** A single pass cannot simultaneously perceive, reason, and write a long faithful review, so it sacrifices and drops $\sim 1/3$ of what it saw. Decomposition gives perception a dedicated externalization pass and synthesis a dedicated, *full-budget* pass over complete text. Collapsing the two into one call crashes coverage or fails to emit a review at all (the transcript exhausts the output budget), so the separation, not the modality, is the active ingredient.

We call the pattern **perceive–externalize–synthesize**. This account also predicts the two failure modes (§5): a compression bottleneck when synthesis cannot fit the transcript in its output budget (Mode A), and a training-prior leak when removing the source lets synthesis confabulate (Mode B). The latter is a genuine but secondary cost of dropping the modality, not the engine of the win.

This relates to, but is distinct from, prior single-modality work. Cuervo et al. [2] and the Cascade Equivalence Hypothesis [3] find speech-LLMs internally behave like ASR→LLM cascades, and Step-Audio-R1 [5] shows audio LLMs reasoning over a *textual surrogate* of the audio. Our activation-level tests find *no* reasoning-substrate penalty for propositional content in *either* modality (§4): a VLM reasons over an image of text as well as over text (E6), and, once the speech is intelligible, an omni model reasons over spoken facts as well as written ones (E7), with an apparent audio gap vanishing when low-quality TTS is replaced by a clean neural voice. The Step-Audio-R1 effect is best read as task-payload-specific, not a contradiction: its tasks (music, environmental sound) have *acoustic* answers that transcription discards. On documents, DocVLM [6] reports OCR-augmented VLMs beating pure-vision VLMs at fixed budget, while vision-only configurations dominate on GUI tasks [9, 10]. Few cross-modality

same-weights measurements exist, and fewer still isolate *why*. This paper provides both, focused on **long-form review generation**, the deployment case where end-to-end models are observably weak (they drop content and embellish) and two-pass decomposition is the obvious alternative.

Findings. (i) Two-pass decomposition (C_1) beats end-to-end (C_0) across 21 Gemini cells on both faithfulness (18/21) and coverage (16/21), both sign-test significant, despite the synthesis pass seeing strictly less (§3). (ii) The gain scales inversely with the C_0 baseline ($r = -0.45$): decomposition recovers most where the monolithic pass satisfies most. (iii) *The mechanism is generation-side, not perception or substrate* (§4): the dropped content sits in the model’s own transcript, text and image-of-text are read equally well (on open VLMs and frontier models alike), yet collapsing the two passes into one call crashes coverage. (iv) The account predicts two failure modes—compression overflow (Mode A) and training-prior leak (Mode B, replicated on Claude). (v) A quote-grounded iterative-refinement variant ($C_{1\text{iter}}$) suppresses Mode B where it binds but trades coverage elsewhere. (vi) GPT-5.4 and Claude judges agree on direction (coverage 19/21, hallucination 14/21; per-probe median 0.94).

Contributions. (1) A 21-cell same-weights measurement of end-to-end (C_0) vs. two-pass decomposition (C_1), across two modalities, three vendors, and two judges. (2) *A nine-experiment mechanistic study* (§4) that adjudicates among perception, substrate, and generation-side accounts, ruling the first two out and supporting **perceive–externalize–synthesize**: perception coverage vs. surfacing (E3), an open-VLM cross-modal logit-lens under load (E6), an omni-model audio analogue (E7), behavioral cross-modal retrieval (E8) and multi-hop-reasoning (E9) probes on the headline frontier models, the single-call collapse test (E2), a both-modalities control (E1), and DPI/native-text controls (E4/E5, reported with their confounds). (3) A two-mode account of where decomposition breaks down (Mode A compression, Mode B training-prior leak) with deployment triggers, derived from the mechanism. (4) An iterative-refinement variant ($C_{1\text{iter}}$) that re-grounds synthesis with verbatim quotes. (5) Systematic robustness checks ($n=5$ multi-seed on four cells, inter-judge on all 21, probe-circularity on 6, reference-bias on 4), with all retract/preserve verdicts reported transparently.

2 Protocol

For each source (audio or scanned paper): (1) A **reference transcript** is produced by Gemini 3.1 Pro ASR/OCR with chunking (30-min audio chunks, 1-page-per-call OCR, Appendix A). (2) C_0 *reason-over-modality* (end-to-end): the model under test reads the raw modality and writes a review article in one call. (3) C_1 *reduce-to-text-then-reason* (the same-weights cascade): the model under test produces its own transcript (Pass-1, same prompt as the reference), then writes the review from that transcript (Pass-2, text-only). The only thing that changes between C_0 and C_1 is whether the reasoning step sees pixels/audio or text. Figure 1 schematizes the two configurations.

Scoring. GPT-5.4 (reasoning_effort=high) acts as judge. From the reference, it extracts 30–50 atomic factual probes. For each review the judge then lists every claim not supported by the reference (*hallucination test*) and marks each probe as COVERED / MISSING (*probe*

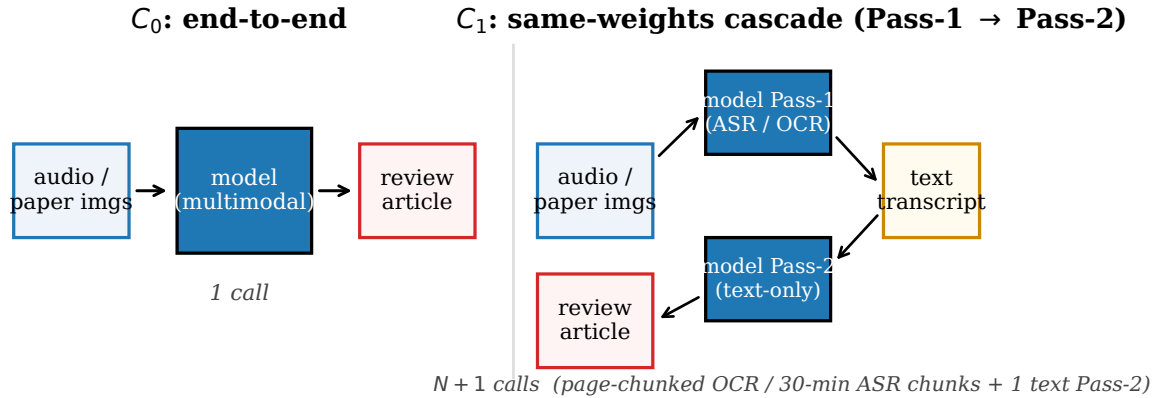


Figure 1. Reason-over-modality (C_0) vs. reduce-to-text-then-reason (C_1), same weights. C_0 is one multimodal call: the model under test reads the raw audio or page images and writes a comprehensive review, reasoning directly over the modality. C_1 first reduces the source to text (Pass-1 ASR/OCR call, same model weights), then reasons over that transcript alone (Pass-2, text-only, with the raw modality absent from the Pass-2 context). The two configurations differ only in the representation the reasoning step operates over.

coverage). The protocol is sharper than short-form QA: the review must be both exhaustive (high coverage) and faithful (zero hallucination). Because most frontier models fail the zero-hallucination bar, we report counts and rates as operational measures.

Noise floor and statistics. We treat $|\Delta h| \leq 1$ on cells with $h < 10$, and $|\Delta \text{cov}| \leq 0.04$ (≈ 2 probes on a 50-probe cell), as within noise (presentation conventions calibrated against inter-judge direction flips, §7). For aggregate effects we report Wilson 95% CIs on coverage rates, percentile-bootstrap 95% CIs ($n_{\text{boot}}=10000$, seeded) on cross-cell means and Pearson r , and sign tests against $\Delta=0$.

Cells. 21 Gemini cells: 11 audio sources (26–85 min talks, lectures, panels, and meetings) and 10 papers (9–176 pp), the latter being seven NASA NTRS reports (1968–1972) plus three modern post-cutoff arXiv reviews; full source list in Appendix A and per-cell counts in Appendix B. *Cross-vendor extensions* reuse the Gemini reference and probes (isolating the question to whether the advantage depends on the C_0/C_1 model): Claude Opus 4.7 (a complete C_0 -vs- C_1 comparison on the six papers where its multimodal C_0 fits, §3; C_1 -only on the longer papers) and Mimo v2-omni on every cell where its C_0 does not return empty.

Notation. C_0 : reason-over-modality (end-to-end). C_1 : reduce-to-text-then-reason (single-pass cascade). Subscript c (C_{1c}): chunked Pass-2 (sub-summarize per chunk, then merge). Superscript concat: chunked, no merge. Superscript strip: citation surface forms replaced by neutral [REF] placeholders before Pass-2. Subscript iter: the iterative-refinement variant of §6.

3 Main result: two-pass decomposition beats end-to-end

Aggregate effect at $n = 21$. The cascade reduces hallucinations on 18/21 cells (mean $\Delta_{\text{halluc}} = -3.86$, bootstrap 95% CI $[-5.62, -2.33]$, sign-test $p = 0.0015$) and improves coverage on 16/21

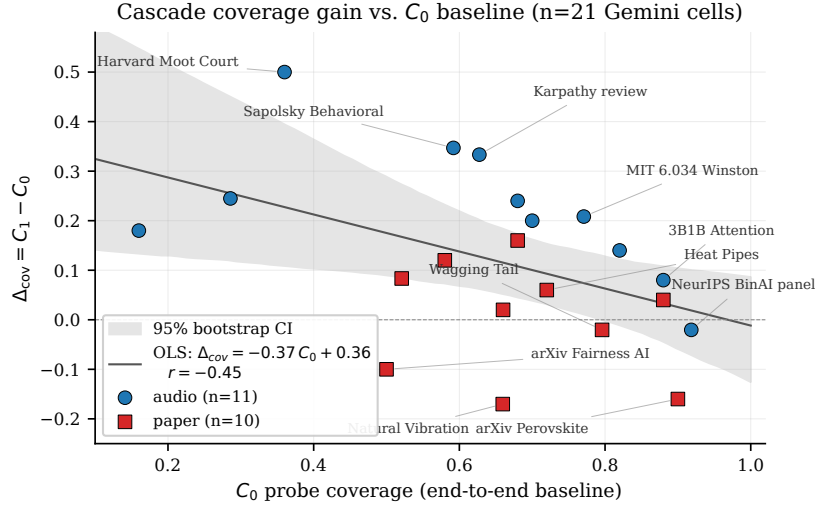


Figure 2. Reduce-to-text coverage gain vs. C_0 baseline across all 21 Gemini cells. The worse the model reasons over the raw modality (lower C_0), the more reducing to text recovers. Circles: audio. Squares: paper. OLS line with bootstrap 95% CI band, Pearson $r = -0.45$ with bootstrap CI $[-0.73, -0.10]$ (fraction $r \geq 0$: 0.9%). The five counter-cells (NeurIPS panel, Wagging Tail, arXiv Perovskite, Natural Vibration, arXiv Fairness AI) sit at high C_0 baseline (no headroom) or trigger Mode A/B (§5).

(mean $\Delta_{\text{cov}} = +0.118$, $[+0.048, +0.189]$, $p = 0.027$); both intervals exclude zero. The effect is robust to median substitution on the four high-count judge-spread cells (§7). Per-cell counts and Wilson CIs are in Appendix B.

Direction of effect by baseline. Coverage gain scales inversely with the end-to-end baseline (Figure 2). This is the central prediction of the reduce-to-text account: C_0 measures how well the model already reasons over the raw modality, and the cells where it reasons worst are exactly where moving the reasoning onto text recovers the most. Excluding the two meeting-minutes re-prompts (artificially low absolute coverage because their probes use atomic-fact style while the prompt asks for structured minutes), $r = -0.47$ on $n = 19$ with CI $[-0.83, +0.10]$. The direction survives, but the CI just crosses zero on this subset.

Cascade-positive cells. The largest coverage gain is Harvard Moot Court audio (+50pp, multi-speaker legal argumentation with little C_0 headroom); the cleanest hallucination win is 3B1B Attention (11h $\rightarrow$ 0h). Gains concentrate on audio and on moderate papers (Appendix B).

Counter-cells. The five cells where cascade fails to improve on at least one axis split into two clusters: *high-baseline / no-headroom* (NeurIPS panel, Wagging Tail, arXiv Perovskite, all $C_0 \geq 0.80$) and *mode-triggered failures* (Natural Vibration, Mode B; arXiv Fairness AI, Mode A; §5). Perovskite’s loss shrinks from -16pp at $n=1$ to -6pp under $n=5$ multi-seed (Appendix F), so we treat it as near-noise-floor.

Cross-vendor: 3/3 vendors confirm cascade on a moderate paper. On Thermal Analysis 66pp (the moderate-length cell where all three vendors realize C_0 and C_1), all three improve under cascade on coverage, and Gemini and Claude both cut hallucinations sharply (Figure 9a).

Despite Claude producing reviews $\approx 2\times$ longer, Gemini and Claude converge to a common ≈ 1.4 hallucinations per 1k output words under cascade from different end-to-end rates (Appendix L).

Within-Claude: a complete C_0 -vs- C_1 comparison on six papers. To address the single-vendor concern beyond the one Thermal cell, we ran Claude Opus 4.7 end-to-end *and* cascade on all six NASA papers short enough for Claude’s multimodal C_0 to fit the request payload (9–104pp, Table 8). Cascade reduces Claude’s hallucinations on 5/6 cells (the lone exception is the Heat Pipes Mode-B cell, §5), and the magnitude follows the same inverse-baseline law as the Gemini headline (Figure 2): the four short/clean papers are already near-ceiling on C_0 , so coverage stays flat, while the one moderate cell with genuine headroom (Thermal) is a clean Pareto win. We do *not* claim independent significance at $n=6$ (sign-test $p=0.22$, dominated by the single Mode-B regression); the value is that the cascade *direction* replicates within a second model family on a complete within-vendor comparison, with magnitude tracking C_0 headroom as the mechanism predicts.

Cross-model audio (Gemini 2.5 Flash), clean and unfounded. Because the only non-Pro audio arm (Mimo) is bitrate-confounded, we ran a full-quality C_0 -vs- C_1 on Gemini 2.5 Flash over five representative audio cells (Appendix D). 2.5 Flash turns out to be a *strong* end-to-end audio reader—its C_0 coverage exceeds Pro’s on four of five cells—so the outcome is headroom-gated exactly as the inverse-baseline law predicts: the cascade wins where C_0 has room (MIT $-3h/+13pp$, Harvard $-7h$) and is flat-to-slightly-negative where C_0 is already saturated (NeurIPS 0.98; Karpathy 0.92, which even regresses on faithfulness), improving faithfulness on 3/5 cells overall. This corroborates the *mechanism* ($\text{gain} \propto \text{headroom}$) rather than a uniform audio win, and it removes the bitrate confound. Gemini 3 Flash cannot serve as a second subject: its end-to-end C_0 runs, but its verbatim Pass-1 transcription is refused by the model’s recitation filter on copyrighted audio, so the cascade’s transcription pass is not realizable on it—another realizability limit alongside Claude’s lack of audio input and the long-paper payload caps.

Long-paper divergence. On longer cells the three vendors diverge sharply (full numbers in Appendix C). On Heat Pipes 104pp, Gemini cascade is small Pareto-positive while Claude cascade *regresses* on faithfulness, leaving C_0 on the Pareto frontier—the Mode-B signature (§5). On SP-5100 176pp both Gemini and Claude C_1 sit at very high hallucination rates with the same bibliographic-expansion signature, and Claude C_0 is not realizable (176 page-images exceed Anthropic’s payload limit even with the 1M-context beta). On the modern arXiv reviews the cascade direction can *reverse*, which we attribute mainly to denser typesetting forcing more aggressive synthesis-pass compression at the same output budget (Mode A). The long-paper picture motivates the mechanism study and the two failure regimes (§4–5).

4 Mechanism: perceive, externalize, synthesize

Why does decomposition win? Three hypotheses are consistent with the headline result but make different predictions. **H1 (substrate):** the model reasons better over text tokens than over raw audio/image tokens, so moving the work onto text helps. **H2 (externalization /**

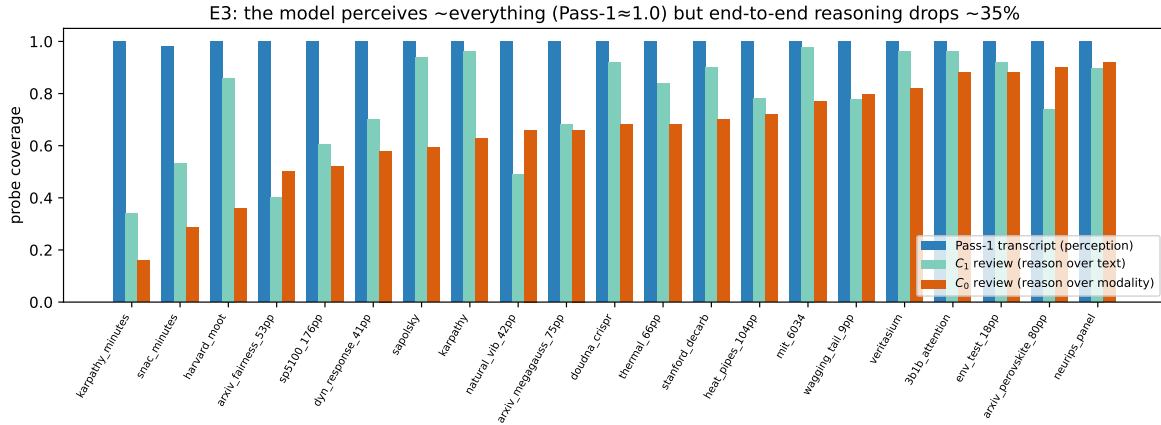


Figure 3. E3: perception is not the bottleneck. Per cell (sorted by C_0 coverage): the model’s Pass-1 transcript (blue) covers ≈ 1.0 of probes everywhere, while the end-to-end C_0 review (orange) surfaces only 0.65 on average and the C_1 synthesis (green) 0.77. Pooled, 361/362 probes that C_0 drops are present in the model’s own transcript.

generation under load): a single pass cannot simultaneously perceive, reason, and write a long faithful review. Decomposition gives perception a dedicated externalization step and synthesis a dedicated, full-budget step over complete text. **H3 (attention dilution):** raw-modality tokens are numerous and low-density, diluting the reasoning attention. We ran nine experiments (E1–E9) to separate them. The verdict is that H2 is the operative mechanism, H1 is *not* supported (and is directly contradicted for documents—including, now, on the headline frontier models themselves, for retrieval (E8) and multi-hop reasoning (E9)), and H3 is untestable on Gemini with our tools. All experiments reuse the exact probe/hallucination judge of §2, and artifacts are released (§N). The two load-bearing generation-side contrasts (E1, E2) are run at $n=5$ seeds on all 21 cells (§4, statistical treatment). E4/E5 remain single-seed controls on an 8-cell subset and are reported as such. §4 closes with the statistical caveats, and §8 with the validation plan.

E3: the bottleneck is not perception. We scored the model’s *own* Pass-1 transcript against the same probes used for the reviews (all 21 cells). The transcript covers a mean 99.9% of probes, far above the C_0 and C_1 reviews (Figure 3). Decisively, of the 362 probes the end-to-end pass drops across the suite, 361 (99.7%, Wilson 95% CI [98.5%, 99.9%]) are present in the model’s own transcript: the content is lost *downstream* of perception. (The transcript ceiling is partly definitional—probes come from a near-identical reference—but the load-bearing inference is one-directional: content C_0 drops is nonetheless capturable-as-text by the same weights.)

E6: text is not an intrinsically better reasoning substrate. We tested H1 directly at the activation level on an open VLM (Qwen2.5-VL-7B), controlling for prior knowledge with novel synthetic facts about nonsense entities (e.g. “the Zorblatt device weighs 7 kilograms”). Each fact is presented either as digital text or as an *image* of that same text, with a matched single-token question, and a layer-wise logit-lens reads off the log-probability and rank of the answer token at every layer. The two modalities are indistinguishable for an isolated fact, and—crucially—*under load*: packing $K \in \{1, 4, 8, 16, 24\}$ facts into one stimulus, image and text stay at 1.00 accuracy with a readout-depth gap < 1 layer (TOST-equivalent within ± 1

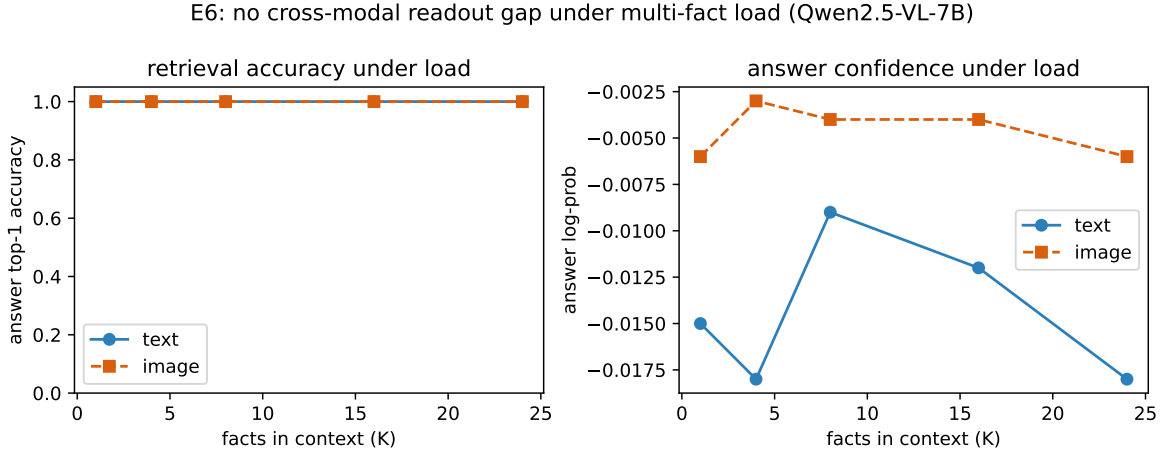


Figure 4. E6: no cross-modal substrate gap, even under load (Qwen2.5-VL-7B logit-lens). Answer accuracy (left) and answer-token log-prob (right) as a function of facts in context K , for content given as text vs as an image of that text. The curves overlap: identical propositional content is read equally well from pixels and from text up to $K=24$. The 3B model and per- K readout depths are in Appendix G.

layer) at every K (Figure 4). The same null replicates on a second VLM (Qwen2.5-VL-3B); the per- K ladder for both models is in Appendix G. For legibly rendered text there is simply no cross-modal reasoning penalty: the strong form of H1 is false for documents. (The 32B variant verbalizes answers in a form our single-token matcher cannot score, so it is excluded; frontier models and dense scanned pages remain for the validation plan.)

E7: no audio substrate gap either, once the speech is intelligible. We replicate the E6 logit-lens on an omni model (Qwen2.5-Omni Thinker), presenting each novel fact as text vs as *synthesized speech*, across three TTS engines of increasing quality (Table 1). The result is a methodological caution turned finding: with low-quality TTS (espeak, only 71% of clips intelligible to the model’s own ASR) audio *appears* to have a substrate gap, but this tracks *intelligibility*, not modality—with two independent clean neural voices (88% and 96% intelligible) the gap vanishes and is TOST-equivalent within ± 1 layer. So neither documents (E6) nor audio (E7) carry an intrinsic reasoning-substrate penalty for propositional content. (The espeak contrast is reported as a cautionary result: garbled TTS manufactures a spurious gap, so audio-substrate claims must control for TTS intelligibility.)

Table 1. E7 audio readout (Qwen2.5-Omni Thinker, 24 novel facts): the apparent audio substrate gap is a TTS-intelligibility artifact. “Intellig.”: fraction of clips the model’s own ASR transcribes correctly. Depth-gap = audio–text readout depth (layers). TOST SESOI ± 1 layer.

TTS	intellig.	text acc	audio acc	depth-gap 90% CI	TOST-equivalent
espeak (robotic)	0.71	0.96	0.83	[0.64, 1.78]	no (spurious gap)
mms-tts-eng (clean)	0.88	0.96	1.00	[0.18, 0.57]	yes
fish-speech-1.5 (natural)	0.96	0.96	1.00	[0.21, 0.54]	yes

E8: the no-substrate-gap result holds on the headline frontier models. E6 and E7 establish the substrate null with a logit-lens, but only on small *open* models (the lens needs internal

activations). A reviewer can fairly object that the mechanism is then established on proxies rather than on the Gemini 3.1 Pro and Claude Opus 4.7 models that produce the headline result. We close that gap with a *behavioral* probe on the exact headline models: pack $K \in \{1, 8, 16, 24, 32, 48\}$ novel nonsense-entity facts (distinct two-digit values, so prior knowledge gives no shortcut) as text or as an *image* of that text, and ask a single-number question about one target entity ($T=20$ seeded trials/cell). On *both* frontier models, legible-image retrieval accuracy equals text accuracy exactly (gap $+0.000$, TOST-equivalent within ± 0.1) at *every* K up to 48 (Table 2), extending the E6 null past its $K=24$ ceiling and onto the frontier weights. The probe is demonstrably *sensitive*: a degraded positive control (small font, downscaled and blurred to low-DPI-scan quality) collapses to 0.15–0.27, decisively non-equivalent. So the legible-content null is a true equivalence, not a task-too-easy ceiling. The contrast with the headline is the point: single-fact retrieval is *perfect* from pixels and text alike, yet C_0 on these same models still drops $\sim 1/3$ of the content (E3)—the loss is generation-side, not substrate, on the frontier model itself.

Table 2. E8 behavioral cross-modal substrate probe on the *headline frontier models* ($T=20$ seeded trials per K , single-number retrieval). “legible img”: image of the same text at 34px. “degraded img”: small-font, downscaled+blurred low-DPI-scan control. Gap = image–text accuracy; TOST SESOI ± 0.1 . Legible image is exactly equivalent to text at every load K on both models; the degraded control is decisively non-equivalent, confirming probe sensitivity.

Model	cond	acc @ $K=1$	acc @ $K=48$	pooled gap (95% CI)	TOST-equiv
Gemini 3.1 Pro	text	1.00	1.00	—	—
	legible img	1.00	1.00	$+0.000 [0.00, 0.00]$	yes
	degraded img	0.15	0.25	$-0.733 [-0.81, -0.65]$	no (control)
Claude Opus 4.7	text	1.00	1.00	—	—
	legible img	1.00	1.00	$+0.000 [0.00, 0.00]$	yes
	degraded img	0.15	0.25	$-0.850 [-0.91, -0.78]$	no (control)

E9: the substrate null extends from retrieval to multi-hop reasoning. E8 equalizes single-fact *retrieval*; a reviewer may fairly note that reasoning is not retrieval. E9 forces composition: the model must pull $M=6$ specified facts out of a K -fact packed set and *sum* them—the total is never a visible number, so a correct answer requires genuine multi-fact reasoning, not retrieve-and-echo—as text vs. legible image vs. the E8 degraded control ($T=20$ /cell, Table 3). On Gemini 3.1 Pro, text and legible image are both at ceiling and TOST-equivalent (gap -0.013). On Claude Opus 4.7—which, run without extended thinking, has genuine *headroom* (text accuracy 0.825, slipping on the in-context sum)—the legible image is *not worse than text*; it is *reliably better* (image 1.00, gap $+0.175$, 95% CI $[0.10, 0.26]$; the 14/80 text errors are all near-miss arithmetic slips that the image condition gets right). This is the decisive direction: the substrate hypothesis H1 predicts image $<$ text under reasoning load, yet with headroom we observe image $\geq$ text. (We read the Claude effect as a within-prompt echo of this paper’s own thesis—reading from an image nudges the model to externalize the numbers before summing—but report only the direction, not a mechanism.) The degraded control collapses both models to ≈ 0 , so the probe would register a modality penalty if one existed. We could not drive clean-legible-*text* accuracy below ceiling on Gemini at any difficulty we tried (sums of up to 12

facts, k th-largest over 32), so the Gemini equivalence is at ceiling while Claude supplies the headroom regime.

Table 3. E9 multi-hop reasoning probe on the headline frontier models: retrieve $M=6$ packed facts and sum them (answer never directly visible), $T=20$ /cell over $K \in \{8, 16, 24, 32\}$. Gap = image–text accuracy; TOST SESOI ± 0.1 . On Gemini both are at ceiling (equivalent); on Claude, where text has headroom, the legible image is *better*, not worse—opposite to the text-is-a-better-substrate prediction. The degraded control floors on both, confirming sensitivity.

Model	text acc	legible-img acc (gap)	TOST-equiv	degraded-img acc
Gemini 3.1 Pro	1.00	0.99 (−0.013)	yes	0.01
Claude Opus 4.7	0.825	1.00 (+0.175)	no (image <i>better</i>)	0.00

E2: the win requires a separate, full-budget synthesis pass. If decomposition merely externalizes content into one context, doing both steps in a single call should suffice. It does not. A single call instructed to first transcribe verbatim and then write the review (modality attached) *collapses*: coverage falls far below C_1 on every paper that produced a review, and on several cells the call never emits a review at all—the transcript consumes the entire output budget (Table 4). This is confirmed at scale: at $n=5$ seeds on all 21 cells, single-call coverage is below C_1 by mean $\Delta_{\text{cov}} = -0.111$ (bootstrap 95% CI $[-0.170, -0.038]$, Wilcoxon $p = 0.019$, mixed-effects $p < 10^{-3}$), with 7 runs emitting no review at all. The active ingredient is the *decomposition into two passes with independent output budgets*, not text-in-context per se.

E1: removing the modality is a minor faithfulness bonus, not the engine. Giving the synthesis pass *both* the transcript and the raw modality (C_{1+}) does not break the benefit and sometimes maximizes coverage (Table 4); consistent with E6, pixels are not harmful to content recovery. What their presence does do is re-introduce a few hallucinations relative to pure-text C_1 , the same effect Mode B names. At $n=5$ seeds on all 21 cells this holds: re-attaching the modality moves coverage by only $\Delta_{\text{cov}} = +0.069$ (95% CI $[-0.001, +0.154]$, inconclusive) while adding hallucinations on 14/21 cells. So dropping the modality buys faithfulness at the margin, but the larger coverage win comes from the pass structure (E2), not from removing pixels.

E4, E5: two controls that did not separate the hypotheses (reported for completeness). E4 sought to vary vision-token count via DPI on a fixed document to probe H3. But Gemini normalizes images internally, so 96 and 200 DPI yield near-identical prompt token counts (112,007 vs 111,855 on Heat Pipes). The manipulation does not move the independent variable, and H3 is untestable this way on Gemini. E5 fed native digital text (pdf to text) to a single text-only call as a clean H1 test, but on two-column arXiv PDFs pdf to text scrambles reading order, so native-text coverage is confounded downward (Fairness AI 0.34, below even C_0). Neither supports H1, so we report both with their confounds rather than omit them.

Synthesis. The evidence converges on **perceive–externalize–synthesize**. The model perceives the content (E3). Pixels-vs-text is not the reasoning bottleneck for legible documents, on open VLMs at the activation level (E6) and on the headline frontier models behaviorally, for both retrieval (E8) and multi-hop reasoning (E9). And the recovery requires decomposing the

Table 4. Mechanistic conditions on the 8-cell subset (hallucinations h / probe-coverage). C_0 : end-to-end (modality). C_1 : two-pass decomposition (text). C_{1+} : synthesis pass given both transcript and modality (E1). single-call: transcribe-then-review in one response (E2, “–” = no review emitted, budget exhausted). native-text: single text-only call from pdf to text (E5, arXiv only). Single-seed.

Cell	C_0 (img/aud)	C_1 (text)	C_{1+} both	single-call	native-text
MIT 6.034 (audio)	17h/0.77	4h/0.98	–	– (budget)	–
3B1B Attention (audio)	11h/0.88	0h/0.96	5h/1.00	1h/0.80	–
Wagging Tail 9pp	3h/0.80	1h/0.78	2h/0.78	5h/0.63	–
Natural Vibration 42pp	4h/0.66	5h/0.49	9h/0.79	– (budget)	–
Thermal 66pp	5h/0.68	2h/0.84	1h/0.68	3h/0.40	–
Heat Pipes 104pp	14h/0.72	11h/0.78	10h/0.66	– (budget)	–
arXiv Fairness 53pp	14h/0.50	15h/0.40	11h/0.38	11h/0.24	13h/0.34
arXiv Perovskite 80pp	2h/0.90	3h/0.74	5h/0.84	5h/0.80	4h/0.80

task into two passes with independent output budgets (E2), not merely removing the modality (E1). The headline inverse correlation ($r = -0.45$) follows directly: a low C_0 baseline marks cells where the monolithic pass satisfies most, and E3 shows the dropped content is sitting in the transcript waiting to be synthesized. The two failure modes (§5) are the account’s two predicted breakdowns: synthesis-budget overflow (Mode A) and source-removal confabulation (Mode B).

Statistical treatment and scope. E3 spans all 21 cells; E6 is a controlled within-item logit-lens on two open VLMs; the generation-side contrasts E1/E2 are multi-seed ($n=5$) on all 21 cells; E8/E9 confirm the substrate null behaviorally on the headline frontier models for retrieval and multi-hop reasoning; and E4/E5 are single-seed controls that did not separate the hypotheses. We therefore treat the mechanism as *strongly supported by E3, E6/E8/E9, and the multi-seed E2, corroborated by E1*, with E4/E5 inconclusive. What remains for replication is breadth, not power: activation-level access on frontier models, real scanned pages, and task generality beyond long-form review (§8).

5 Where decomposition breaks down: two failure modes

The perceive–externalize–synthesize account (§4) predicts exactly two ways decomposition can fail, and both appear in the data: when the synthesis pass cannot carry the externalized transcript through its output budget (Mode A), and when removing the raw source lets the synthesis pass fill gaps from prior knowledge (Mode B, the secondary cost of dropping the modality). These are the two modes below.

Mode A: compression bottleneck (3 papers). On long sources, Pass-2 has a fixed output budget ($\approx 2k$ words). When the cascade transcript is much longer than what fits, Pass-2 must compress, dropping content the same way C_0 does. Three Gemini cells exhibit this signature: Heat Pipes 104pp (Pass-1 transcript 22k words, compression ratio $\sim 11\times$, C_1 gain +6pp), arXiv Fairness AI 53pp ($\sim 15\times$, which *loses* –10pp), and arXiv Perovskite 80pp ($\sim 9\times$, –16pp at $n=1$, –6pp under multi-seed). The fix tested on Heat Pipes is **chunked Pass-2**: split the transcript into $\sim 5k$ -word chunks, summarize each, then concatenate (or merge).

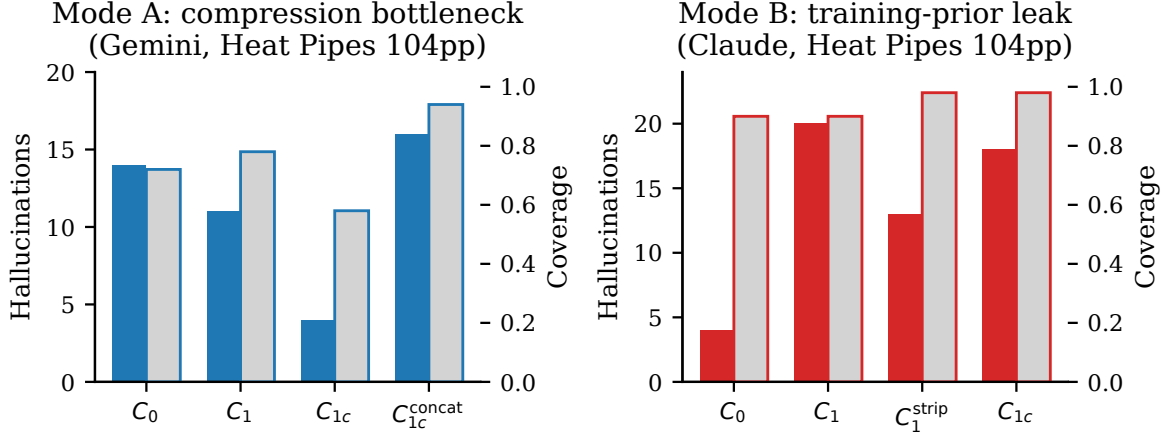


Figure 5. Two cascade failure modes on Heat Pipes 104pp. *Left:* Mode A (Gemini): Pass-2 must compress 22k $\rightarrow$ 1.8k words. Chunked Pass-2 (C_{1c}^{concat}) relieves the bottleneck and lifts coverage 0.78 $\rightarrow$ 0.94 at +5 halluc cost (a trade along the frontier). *Right:* Mode B (Claude): C_1 hallucinates several-fold more than C_0 (4 $\rightarrow$ 20 on the seed plotted, $n=5$ Pass-2 mean 13, range 9–20, Appendix H). Citation-stripping (C_1^{strip}) reduces hallucinations 20 $\rightarrow$ 13 at $n=1$ (the $n=5$ replication retracts the magnitude, §5 and Appendix E). Halluc on left axis (lower better), coverage on right axis (higher better).

Heat Pipes Gemini C_{1c}^{concat} lifts coverage 0.78 $\rightarrow$ 0.94 (+16pp) at +5 halluc (Figure 5 left). At $n=5$ multi-seed verification (Appendix E, Table 13) the chunked-Pass-2 direction replicates on Fairness AI and Perovskite but *not* on Megagauss, so we report chunked Pass-2 as a 2/3 replicated Mode-A fix.

Mode B: training-prior leak (4 papers). This is the genuine cost of reduction: removing the raw source removes a grounding anchor, and on documents with heavy training-data overlap Pass-2 fills the gap from prior knowledge when the transcript abbreviates a detail. The Heat Pipes paper cites foundational references like “Reference 2 (Cotter)”; Claude’s C_1 expands these to “Cotter’s 1965 analysis (LA-3246-MS)” —the year and report number are training-data details, not in the transcript (examples in Appendix M). The same bibliographic-expansion signature appears on all four long citation-heavy Claude cells (Heat Pipes, SP-5100, Fairness AI, Megagauss), whereas the moderate citation-light Thermal 66pp paper shows only 4 Claude C_1 hallucinations.

Two diagnostic signals. Two thresholds emerge as starting points for practitioner triage:

- **Mode A trigger:** Pass-1 transcript length $\gtrsim 5 \times$ Pass-2 output budget with C_1 coverage gain $< +10\text{pp}$ (Heat Pipes Gemini: 11 $\times$, +6pp).
- **Mode B trigger:** input transcript contains $> \sim 2$ “Reference N ” / “Ref. N ” surface forms per 1k words *and* the topic appears in widely cited pre-cutoff literature. Heat Pipes 104pp has 55 such surface forms (2.48/1k). Thermal 66pp has 0 and shows no Mode B even on Claude.

The two modes can compound (Mimo Heat Pipes shows both) or operate in isolation (Heat Pipes Gemini is dominantly Mode A, Heat Pipes Claude dominantly Mode B).

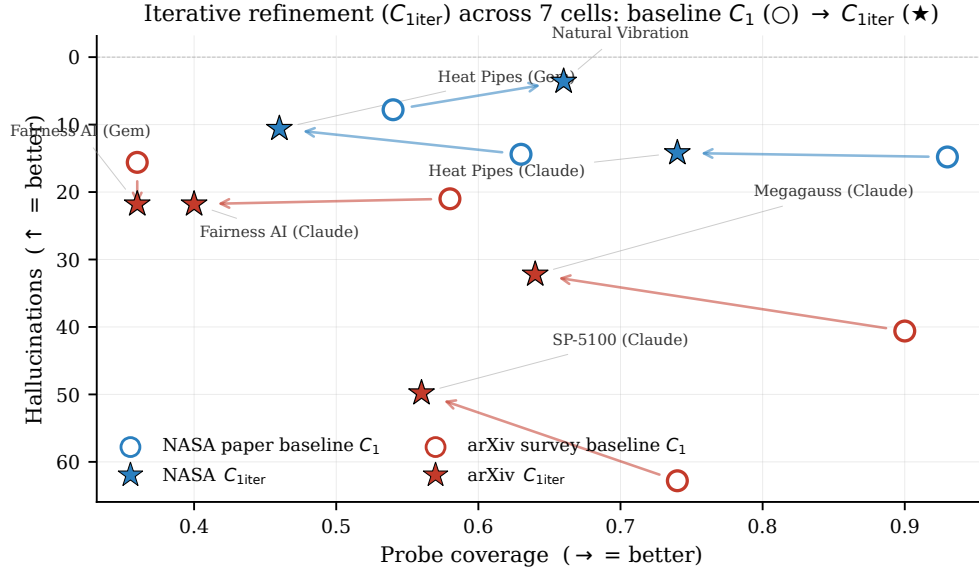


Figure 6. Iterative-refinement Pareto frontier across 7 cells ($n=5$ means). Each cell’s plain C_1 baseline (open circle) is connected by an arrow to its C_{1iter} result (filled star). Blue: NASA papers. Red: modern arXiv surveys. Only Natural Vibration moves down-and-right (fewer hallucinations *and* more coverage, a Pareto win). The other long cells move left (coverage cost from the rewrite budget), with hallucinations falling on the high-count Claude surveys (SP-5100, Megagauss) and rising on arXiv Fairness AI.

Engineering interventions tested. We evaluated two interventions across cells: *chunked Pass-2* (Mode-A target, 2/3 replication under $n=5$ multi-seed, Appendix E) and *citation stripping* (C_1^{strip} , Mode-B target). **Citation stripping does not survive replication:** at $n=5$ Pass-2 calls on Heat Pipes Claude the strip effect on hallucinations is $\hat{\Delta} = -1.8 \pm 2.6$ (95% CI $[-6.9, +3.3]$), the original “20 → 13” point estimate having been within the combined Pass-2 + judge noise envelope (Appendix H). **We retract the citation-stripping claim on Claude**, motivating the stronger structural intervention in §6.

6 Iterative refinement: a quote-grounded Mode-B fix (C_{1iter})

Pipeline. C_{1iter} is a three-step cascade variant:

1. **Pass-2a (extract).** The Pass-1 transcript is fed to the Pass-2 model with a prompt that asks for 50–150 atomic factual claims, each accompanied by a verbatim quote (≤ 25 words) from the transcript. The prompt explicitly forbids claims without a quotable span.
2. **Faithfulness filter.** Each (claim, quote) pair is string-matched against the Pass-1 transcript on a normalized form (lowercased, punctuation collapsed). Exact substring passes; otherwise a sliding-window fuzzy match at a 0.85 similarity threshold accepts close paraphrases (handling minor OCR/ASR substitutions). Claims whose quote fails are dropped.
3. **Pass-2b (rewrite).** The surviving claims are fed back as a bullet list to a final text-only rewrite call that produces the review using *only* the listed claims, with explicit instructions to add no further facts or attributions.

Filter pass-through is high, and the discipline is in extraction. Across the 7 cells at $n=5$ (mean 83–132 extracted claims per cell), 99.8% of claims (3731/3738) pass the quote filter, with only 7 rejected. Asked for a verbatim quote, the model largely self-disciplines and stops emitting claims it cannot ground, exactly the bibliographic expansions Mode B catalogues. The filter is a tripwire that confirms extraction-step discipline rather than a post-hoc culler.

Table 5. Iterative refinement (C_{liter}) on 7 cells, $n=5$ seeds at $T=0.7$ (hallucinations: mean $\pm$ sd, coverage: mean). Δ columns are vs. that cell’s plain C_1 baseline, re-measured at the same $n=5$. “Pareto”: both axes improve beyond noise. “Trade”: faithfulness gain at a coverage cost. “Net-neg”: no faithfulness gain *and* a coverage loss (or both axes worse). Hallucination counts on the high-count Claude cells (SP-5100, Megagauss) carry large GPT-5.4 judge variance (§7).

Cell (vendor)	C_1 baseline	C_{liter}	Δ_{halluc}	Δ_{cov}	Verdict
Natural Vibration (Gemini)	7.8 $\pm$ 2.2 / 0.54	3.6 $\pm$ 1.4 / 0.66	−4.2	+0.13	Pareto
Heat Pipes (Gemini)	14.4 $\pm$ 2.6 / 0.63	10.6 $\pm$ 3.1 / 0.46	−3.8	−0.17	trade
Heat Pipes (Claude)	14.8 $\pm$ 3.5 / 0.93	14.2 $\pm$ 4.4 / 0.74	−0.6	−0.20	net-neg
arXiv Fairness AI (Gemini)	15.6 $\pm$ 6.6 / 0.36	21.8 $\pm$ 1.0 / 0.36	+6.2	+0.01	net-neg
arXiv Fairness AI (Claude)	21.0 $\pm$ 3.9 / 0.58	21.8 $\pm$ 3.2 / 0.40	+0.8	−0.17	net-neg
SP-5100 (Claude)	62.8 $\pm$ 23 / 0.74	49.8 $\pm$ 13 / 0.56	−13.0	−0.18	trade
arXiv Megagauss (Claude)	40.6 $\pm$ 4.6 / 0.90	32.2 $\pm$ 8.5 / 0.64	−8.4	−0.25	trade

Reading. At $n=5$ (Table 5, Figure 6) the picture is graded: C_{liter} reduces hallucinations on five of seven cells but lowers coverage on every cell except Natural Vibration. It is a clean *Pareto win* only there (short enough that the rewrite does not over-compress); a *faithfulness-for-coverage trade* on Heat Pipes Gemini, SP-5100, and Megagauss (the rewrite re-grounds claims while compressing 30–47k-word transcripts into a ~ 2 k-word budget); and *net-negative* where Mode-A compression dominates (Fairness AI) or the Mode-B gain washes out at $n=5$ (Heat Pipes Claude, where the single-seed faithfulness gain vanishes, echoing the citation-strip retraction of §5).

Mechanism. C_{liter} targets Mode B, since a verbatim-quote requirement rules out unsupported expansions, but it does not relieve Mode A: the rewrite step itself compresses, so coverage falls on every long cell. The net verdict is the balance of recovered faithfulness against that compression cost. It is Pareto-positive only when the source is short enough for the rewrite to re-emit it (Natural Vibration). It is a trade when Mode-B leak is heavy enough to pay for the coverage cost (Heat Pipes Gemini, SP-5100, Megagauss). And it is net-negative when Mode-A compression dominates or the Mode-B gain washes out at $n=5$ (Fairness AI, Heat Pipes Claude). We claim it suppresses Mode B where Mode B binds, not that it is universal.

Connection to retrieval-grounded generation. The pipeline is structurally analogous to retrieval-augmented generation but operates entirely *within* the cascade: “retrieval” is the Pass-1 transcript and “attribution” is the verbatim quote. Compared to citation stripping (§5, retracted), C_{liter} targets the same Mode-B failure with a stronger structural constraint: stripping anonymizes the input citation surface form whereas C_{liter} requires the output claim to be quote-attributable. The trade-off is the rewrite-step compression cost, which is why efficacy is conditional on the cell’s dominant failure mode.

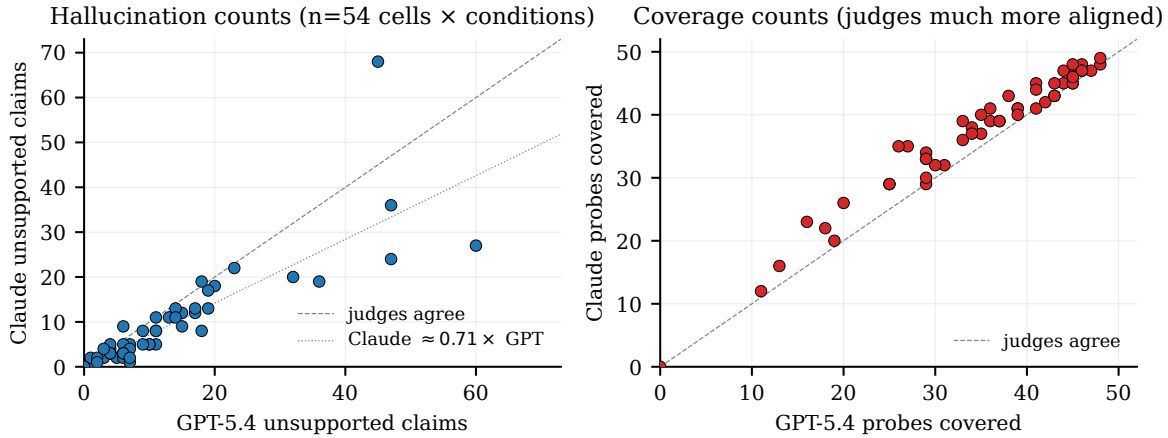


Figure 7. Inter-judge agreement across the 21 Gemini cells plus the Thermal-Claude cross-vendor cell (22 cell-condition pairs, GPT-5.4 vs Claude as judge). Dashed: identity ($y = x$). Dotted: Claude $\approx 0.71 \times$ GPT-5.4 (the median per-pair ratio). Hallucination counts (left) show Claude systematically reporting fewer unsupported claims than GPT-5.4 with most pairs along the $0.71 \times$ line. The two outliers (top-left) are the SP-5100 and Megagauss high-count cells. Probe-coverage counts (right) sit nearly on identity, consistent with per-probe agreement median 0.94.

7 Robustness

We ran four independent robustness checks. Full tables are in Appendices I and J.

Inter-judge replication on all 21 Gemini cells. Claude Opus 4.7 re-scored every C_0 and C_1 review using GPT-5.4’s prompts (Figure 7). Direction agreement is **19/21 on coverage** and **14/21 on hallucination**, with per-probe agreement median 0.94. Disagreements concentrate on near-tie cells (a ± 1 judge difference flips direction) and high-count cells (the judges agree “many claims are unsupported” but differ in absolute count by 10+). Claude systematically reports fewer hallucinations than GPT-5.4 (median ratio 0.71), so the Mode B finding holds under both judges at different absolute counts (Appendix I).

Probe-circularity check (6 cells). Probes are originally extracted by GPT-5.4, which also scores them. We re-extracted probes via Claude on six cells spanning the cascade-direction spectrum (largest audio win, near-noise paper, Mode-A and Mode-B counter-cells, high-baseline saturated, low-baseline long-paper) and re-scored the existing C_0/C_1 reviews. Under *GPT-5.4 judge*, the cascade Δ_{cov} sign is preserved on **6/6** (magnitude shifts ≤ 0.10 pp on all). Under *Claude judge* the sign flips on 1/6 (Fairness AI $-0.10 \rightarrow +0.04$, within noise). The Mode-A loss on Fairness AI and Mode-B loss on Natural Vibration both survive both substitutions.

Reference-bias check (4 cells). The Gemini reference advantages models that share its transcription priors. We re-transcribed two audio cells with Whisper-base and re-OCR’d two paper cells with EasyOCR, re-extracted probes via Claude, and re-judged with both judges (Appendix J). Cascade-coverage direction is preserved on 4/4 under both judges, and cascade-hallucination direction is preserved everywhere except Thermal Analysis under EasyOCR—a

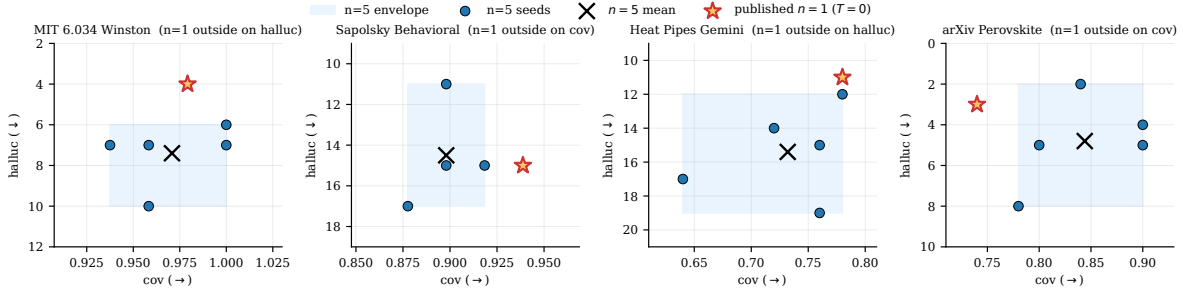


Figure 8. $n=5$ Gemini Pass-2 multi-seed envelopes for the four headline cells. Blue dots: five seeds at $T=0.7$. Black $\times$: seed mean. Orange star: published $n=1$ point at $T=0$ (red edge if outside envelope). Shaded blue rectangle: seed range. On wins (MIT 6.034, Heat Pipes Gemini) the $n=1$ value sits at the low-hallucination edge so the win direction is preserved. On Perovskite the $n=1$ is at the low-coverage edge: cascade-cov loss shrinks from -0.16 to -0.06 .

reference-side artifact, since EasyOCR’s transcript is $\approx 21\%$ shorter and loses figure captions, so claims the cascade correctly reproduces read as unsupported against the noisier reference.

Multi-seed Pass-2 envelopes on four headline cells. We re-ran Gemini C_1 Pass-2 with five seeds at $T=0.7$ on four headline cells (Figure 8, Appendix F). The three cells reported as wins all preserve their direction. The one exception is Perovskite, whose published $n=1$ coverage sits below all five seeds: its cascade-coverage loss shrinks from -16pp at $n=1$ to -6pp under multi-seed, below the per-cell noise floor (hence the near-noise-floor reclassification, §3).

Three sources of stochasticity. Re-runs separate three independent noise sources (full decomposition in Appendix H): Pass-2 generation variance (Claude deprecates temperature, so determinism is not SDK-enforceable; Gemini at $T=0$ is mostly but not strictly deterministic), GPT-5.4 judge variance (essentially deterministic on low-count cells but $\pm 10\text{--}25$ on high-count cells, where long unsupported-claim lists run near the judge’s completion budget), and the consistent observation that *coverage is far more stable than hallucination counts* across all three. Two implications: headline aggregates use median substitution on the four high-count cells, and per-cell hallucination counts > 30 carry $\pm 10\text{--}25$ judge-side uncertainty; and the Mode B *observation* survives all three noise sources while the Mode B *fix* (citation stripping) does not, which motivated $C_{1\text{iter}}$ (§6).

A simple held-out predictor (LOOCV, $n=21$). A leave-one-out logistic-regression predictor on four cell-level features (C_0 probe coverage, $\log_{10}$ source words, C_0 unsupported-claim count, and a paper/audio dummy) reaches LOOCV AUC = 0.65 on $\text{sign}(\Delta_{\text{cov}} > 0)$. All four coefficient signs match the qualitative picture (higher baseline reduces cascade benefit, longer source reduces benefit, more C_0 hallucinations *increases* benefit, paper modality harder than audio). AUC 0.65 is meaningful but well below deployment-ready, so we report it as an honest $n=21$ baseline. Adding a citation-density feature drops AUC to 0.61: the residual Mode-B structure is not captured by a single density feature at this sample size.

8 Discussion

Cascade cost. C_0 is one multimodal call; C_1 is $N + 1$ calls (chunked OCR/ASR plus one text Pass-2), with call-count ratios ranging from $\approx 2\times$ on short audio to $\approx 177\times$ on the 176pp paper, and output-token volume scaling with source length. The 21-cell evidence says cascade pays off on long-form review where end-to-end has measurable headroom, but buys nothing on already-saturated cells (full cost-vs-gain scatter in Appendix K).

Practitioner checklist. Table 6 compresses the recommendations from our cells.

Table 6. Deployment checklist drawn from this paper’s cells.

Source / model class	Diagnosis	Recommended action
Audio talk or moderate paper (40–70pp) on a capable model	Cascade strongly Pareto-positive.	Default C_1 .
Moderate paper with Mode-B Gemini signature (Natural Vibration 42pp)	Mode B dominant, and the cell flips to Pareto-positive under $C_{1\text{iter}}$.	$C_{1\text{iter}}$.
Long paper, Mode A dominant (Heat Pipes 104pp Gemini)	Compression bottleneck.	C_{1c}^{concat} (chunked, no merge).
Long paper, Mode B dominant (Heat Pipes Claude)	Citation-prior leak, but the $C_{1\text{iter}}$ faithfulness gain washes out at $n=5$.	C_0 if it fits. $C_{1\text{iter}}$ buys only a coverage loss.
Long arXiv survey (Fairness AI, SP-5100, Megagauss)	Mode A + Mode B compound. At $n=5$, $C_{1\text{iter}}$ is net-negative on Fairness AI but a faithfulness/coverage trade on SP-5100/Megagauss. Chunked Pass-2 partially replicates.	Prefer C_0 when it fits. If cascade required, use C_{1c}^{concat} with multi-seed averaging.

Limitations. *Cross-vendor n .* We now have a complete within-Claude C_0 -vs- C_1 comparison on six papers (9–104pp, Table 8), on top of the 3/3-vendor Thermal cell. The cascade faithfulness direction replicates within Claude on 5/6, but the four short NASA papers are near-ceiling on C_0 , so the comparison is not independently significant at $n=6$, and on papers longer than 104pp Claude C_0 exceeds Anthropic’s per-request payload cap and remains unrealizable. The single-vendor strength of the headline statistic is therefore reduced but not eliminated. *Judge variance.* The inter-judge replication does not address single-judge intra-rater variance from re-runs with different random seeds. *Probe and reference bias.* Both are partially controlled (6 cells / 4 cells), and the cascade-hallucination direction on paper cells is partly bounded above by Gemini OCR’s recall. *$C_{1\text{iter}}$ judge variance.* $C_{1\text{iter}}$ is verified at $n=5$ (§6), but its two high-count Claude cells (SP-5100, Megagauss) inherit the ± 10 –25 GPT-5.4 judge variance, so their trade verdicts rest on means with wide per-seed spread. *Mimo audio bitrate.* Mimo audio cells ran at 16kbps mono (12kbps for Sapolsky) to fit OpenRouter’s 10MB limit, while the Gemini reference used 32kbps, so Mimo cells partly conflate audio quality with model weakness. A clean, full-quality Gemini 2.5 Flash audio comparison (Appendix D) removes this confound and shows the cross-model audio effect is headroom-gated (a win where C_0 is weak, flat-to-negative where C_0 is already strong), consistent with the mechanism but not

a uniform replication. *Task scope.* All measurements are on long-form review, so we make no claim about short-form QA, code, or agentic tasks. *Mechanism n.* E1/E2 are multi-seed ($n=5$) on all 21 cells, E6 is confirmed on two open VLMs (3B, 7B) on clean rendered text, and the substrate null is now also confirmed *behaviorally on the headline frontier models* for retrieval (E8, K up to 48) and multi-hop reasoning (E9). Two honest caveats remain. First, E8/E9 are behavioral, not logit-lenses, since frontier activations are inaccessible. Second, the frontier models are so capable that clean-legible-text accuracy stays at ceiling on Gemini at every difficulty we tried, so the Gemini reasoning equivalence (E9) is at ceiling; only Claude (run without extended thinking) gave a sub-ceiling regime, where the legible image was *better* than text—directionally against H1, but a single-model headroom point. Real scanned/degraded pages also remain (E8/E9’s degraded control is synthetic), and E4/E5 are single-seed controls (§4). The mechanism rests on E3 (all 21 cells), the multi-seed E2 collapse, and the E6/E8/E9 substrate null. The breadth gaps await the replication below.

Validation plan. The load-bearing generation-side effects (E1/E2) are already multi-seed on all 21 cells, the E6 substrate null is confirmed on two open VLMs, and E8/E9 now confirm it *behaviorally on the headline frontier models* (retrieval and multi-hop reasoning), so the remaining replication is about *breadth and external validity, not statistical power*. We scope it as follows. (1) *Model generality.* Extend the E6 logit-lens beyond the Qwen2.5-VL 3B/7B pair to other open families (InternVL, Llama-Vision) and, where activations become accessible, lift the E8/E9 behavioral frontier confirmation to the activation level on frontier VLMs. Extend E7 across more omni models and higher-quality voices (e.g. CosyVoice2, whose packaged build was unusable in our environment) and push E7 into the multi-fact load regime, since E6 holds for documents under load but audio-under-load is still untested. (2) *Realistic-page stress.* Re-test E6 on real scanned pages (dense multi-column, equations, figures) and degraded scans, where a substrate/dilution effect is most likely to appear, since E4’s DPI knob is inert on Gemini. (3) *Task generality.* Extend beyond long-form review to other generation-under-load tasks (long-document QA with many required facts, multi-document synthesis, meeting-minutes) and to a short-form control where the account predicts *no* decomposition benefit (the bottleneck only binds when output demand is high). (4) *Causal budget manipulation.* Hold the pipeline fixed and vary only the synthesis output budget to confirm Mode A is budget-bound, and ablate transcript completeness to map the coverage–faithfulness frontier.

9 Related work

Speech and documents. Prior cascade-vs-end-to-end results are often read as substrate claims. Our mechanism study suggests they are at least partly about decomposition and modality-specific recoverability. Cuervo et al. [2] document a text/speech understanding gap, and the Cascade Equivalence Hypothesis [3] shows speech-LLMs *internally* behave like ASR→LLM cascades, while X-Talk [4] finds explicitly cascaded speech-to-speech pipelines competitive. On documents, DocVLM [6] reports OCR-augmented VLMs beating pure-vision VLMs on DocVQA, while the OCR-Reasoning Benchmark [7] shows pure-OCR cascades cap below 50% where spatial layout carries meaning, ColPali [8] shows vision-only retrieval beats OCR, and GUI agents do better from raw pixels than from structured-text intermediates [9, 10]. The common thread is not “text is a better substrate” (our E6 finds no such gap for

legible documents) but that decomposition helps when the task is generation-under-load and the content survives transcription, and hurts when the answer lives in layout/signal that transcription discards.

Step-Audio-R1: a contrast, not a corroboration. Step-Audio-R1 [5] is the sharpest statement of the boundary, and it is worth being precise, because it cuts against a naive reading of our claim. They observe that audio LLMs, when asked to deliberate, spontaneously reason over a *textual surrogate* of the audio (e.g. attributing musical sadness to “lyrics mention sadness” rather than to a minor key), and they treat this as the *failure* their modality-grounded reasoning distillation must correct. That is consistent with us, not contradictory: their tasks (music, environmental sound, paralinguistics) are precisely the case where the answer is in the acoustic signal, so reducing to text discards it. We study the complementary regime, long-form review, where the payload is what was *said*. Crucially, our activation tests find no reasoning-substrate gap for propositional content in *either* modality (E6 documents, E7 audio). The Step-Audio-R1 effect is therefore best read as *task-payload-specific*, since their tasks (music, environmental sound, paralinguistics) have answers that live in the acoustic signal, which transcription discards, rather than as a general claim that audio is a worse reasoning substrate. When the payload is propositional, as in long-form review, reducing to text loses nothing and the win is decomposition. Our Mode B (the synthesis pass confabulates once the source is removed) is the within-regime echo of their concern.

Cascade theory and faithfulness. Bengio’s Consciousness Prior [11] argues for sparse discrete bottlenecks between perception and reasoning, and our externalized text bottleneck is a deployment-time analogue. Discrete JEPa [12] makes a related architectural argument within vision. Mode B, where the model embellishes a citation surface form from training-time knowledge, is a self-cascade analogue of the unfaithfulness problem that motivates RAG [13], RARR [14], and Attribution to Identified Sources [15]. Our $C_{1\text{iter}}$ variant adopts the in-context-attribution discipline of RARR but operates entirely within a same-weights self-cascade without external retrieval.

10 Conclusion

On long-form review, a same-weights two-pass decomposition beats end-to-end at $n=21$ on both faithfulness and coverage, even though the synthesis pass sees strictly less, and the benefit scales with how badly the monolithic pass satisfices ($r = -0.45$). The result survives median substitution, probe re-extraction, reference re-generation, and multi-seed Pass-2. Our mechanistic study locates the cause: not perception (99.7% of dropped content is in the model’s own transcript) and not a text-vs-image substrate gap (no logit-lens gap on open VLMs, no behavioral gap on the headline frontier models), but **generation under load**, where one pass cannot perceive, reason, and write a long faithful review at once, and only a decomposition with independent output budgets recovers the content. The boundary is the account’s two predicted breakdowns: synthesis-budget overflow (Mode A) and source-removal confabulation (Mode B), the latter targeted by the quote-grounded $C_{1\text{iter}}$ variant. The practical rule is **perceive, externalize, then synthesize**. The win is the decomposition, not the modality—a deployment-time pattern with predictable failure modes, not a universal architecture.

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A Protocol details

Reference quality. Gemini 3.1 Pro is used for OCR/ASR with chunking. Single-call OCR on a 104-page paper produced a degenerate TOC dot-leader loop (113KB output, 56k “words” but only 264 real-word tokens, 6 page markers out of 104). Chunking (1 page per call, max 8192 output tokens) eliminates this failure mode. For audio we chunk into 30-min segments with 5-second overlap. We discovered a silent server-side truncation on one audio chunk (chunk 1 of IETF SNAC produced 1981 words instead of the expected ≈ 4368 , ending mid-sentence with “Uh, well, see, yeah, that’”). The protocol grew an automatic chunk-level word-count parity check after this incident.

Sources. Audio: Karpathy “State of GPT” (42 min, 2023), MIT 6.034 L1 Patrick Winston (47 min, 2010), Sapolsky Behavioral Genetics 1 Stanford (85 min, post-2026 reupload), Stanford Decarbonized Power Sector Zach Ming (72 min, 2025), Doudna CRISPR Basics (48 min, 2017), 3Blue1Brown “Attention in Transformers” (26 min, 2024), Veritasium “Strange Math” (32 min, 2025), NeurIPS Black-in-AI panel (31 min, 2021), Harvard Law School Ames Moot Court (82 min, 2025), IETF SNAC WG interim 2026-04-16 (60 min, ≈ 5 speakers, recorded post the most likely training-data cutoff). Papers: NASA NTRS IDs 19720009221 (Wagging Tail, 9pp), 19680000724 (Env Test, 18pp), 19690008405 (Dynamic Response, 41pp), 19690013408 (Natural Vibration, 42pp), 19700023812 (Thermal Analysis, 66pp), 19700025120 (Heat Pipes, 104pp), and SP-5100 (Shock and Vibration Technology, 176pp). The arXiv papers are arXiv:2605.09852 (Fairness AI, 53pp), arXiv:2605.13991 (Perovskite Tandem PV, 80pp), and arXiv:2605.11379 (Megagauss Physics, 75pp). NASA papers are rendered as page images at 200 DPI, and arXiv papers at 200 DPI full color.

Models and snapshots. The models are Gemini 3.1 Pro Preview (gemini-3.1-pro-preview), Claude Opus 4.7 (claude-opus-4-7), and Mimo v2-omni (xiaomi/mimo-v2-omni via Open-

Router). The judge is GPT-5.4 (reasoning_effort=high, completion budget 24k tokens), routed via OpenRouter due to scope restrictions on the local OpenAI key.

Mimo audio constraint. OpenRouter’s Mimo provider enforces a 10MB base64 audio input limit. Karpathy required re-encoding from 32kbps to 16kbps mono mp3 to fit, and Sapolsky at 85 min required 12kbps. The Mimo audio Karpathy result (C_0 4h/0.569, C_1 4h/0.588 across 51 probes, 1509 / 1268 review words) is within noise across the $C_0 \rightarrow C_1$ transition, so we treat it as a wash. *gpt-audio* (via OpenRouter) accepted audio input but refused text-only Pass-2 with HTTP 400 “This model requires that either input content or output modality contain audio”. The same-weights cascade is not realizable on this model, and C_0 alone scored 10h/0.765 on Karpathy.

Pareto terminology. We call a cascade condition *strictly Pareto-positive* relative to C_0 when it has both fewer hallucinations and higher coverage by amounts exceeding the noise floor. *Pareto-positive within noise* means strictly better on one axis and no worse than noise on the other. *Pareto-dominant* on a set means no other condition has lower hallucination at equal or higher coverage. These are presentation labels, not statistical claims.

B All 21 cells: raw counts and Wilson 95% CIs

Table 7. All 21 Gemini base cells. Halluc. is the raw unsupported-claim count (single judge call unless noted, with medians across multi-judge data given in brackets for high-count cells). Cov. is k/n (covered probes out of n) with Wilson 95% CI. Cells where C_1 *does not* strictly improve on at least one axis above the noise floor are marked with †. *cite/1k* = count of citation surface forms (numeric brackets + “Ref. N ” word forms) per 1k Pass-1 transcript words.

Cell	len	C_0 halluc	C_0 cov [95% CI]	C_1 halluc	C_1 cov [95% CI]	Δ_{cov}	cite/1k
<i>Audio (11 cells)</i>							
Karpathy review	42m	13	0.627 [0.490, 0.747]	3	0.961 [0.868, 0.989]	+0.333	0.00
Karpathy mtg-min	42m	4	0.160 [0.083, 0.285]	2	0.340 [0.224, 0.478]	+0.180	0.00
SNAC mtg-min	60m	14	0.286 [0.178, 0.424]	11	0.531 [0.394, 0.663]	+0.245	0.00
3B1B Attention	26m	11	0.880 [0.762, 0.944]	0	0.960 [0.864, 0.989]	+0.080	0.00
Veritasium Math	32m	14	0.820 [0.692, 0.903]	6	0.960 [0.864, 0.989]	+0.140	0.00
MIT 6.034 Winston	47m	17	0.771 [0.633, 0.866]	4	0.979 [0.890, 0.996]	+0.208	0.00
NeurIPS BinAI panel	31m	6	0.918 [0.803, 0.969]	4	0.898 [0.778, 0.957]	−0.020†	0.00
Doudna CRISPR	48m	6	0.680 [0.542, 0.792]	3	0.920 [0.812, 0.968]	+0.240	0.00
Stanford Decarb	72m	11	0.700 [0.562, 0.809]	6	0.900 [0.786, 0.957]	+0.200	0.00
Harvard Moot Court	82m	11	0.360 [0.241, 0.499]	4	0.860 [0.736, 0.932]	+0.500	0.00
Sapolsky Behavioral	85m	19	0.592 [0.450, 0.720]	15	0.939 [0.834, 0.980]	+0.347	0.00
<i>Paper, NASA NTRS 1968–1972</i>							
Wagging Tail	9p	3	0.796 [0.664, 0.885]	1	0.776 [0.641, 0.870]	−0.020†	1.42
Env Test	18p	5	0.880 [0.762, 0.944]	4	0.920 [0.812, 0.968]	+0.040†	0.00
Dynamic Response	41p	3	0.580 [0.442, 0.706]	2	0.700 [0.562, 0.809]	+0.120	1.19
Natural Vibration	42p	4	0.660 [0.517, 0.778]	5	0.489 [0.353, 0.628]	−0.170†	6.33
Thermal Analysis	66p	5	0.680 [0.542, 0.792]	2	0.840 [0.715, 0.917]	+0.160	0.36
Heat Pipes	104p	14	0.720 [0.583, 0.825]	11 [med 12, {11, 12, 17}]	0.780 [0.648, 0.872]	+0.060	2.80
SP-5100 Shock	176p	47 [med 46.5]	0.521 [0.383, 0.655]	45 [med 45, {25, 45, 59}]	0.604 [0.463, 0.730]	+0.083	4.00
<i>Paper, arXiv 2026 (post-cutoff)</i>							
arXiv Fairness AI	53p	14	0.500 [0.366, 0.634]	15	0.400 [0.276, 0.538]	−0.100†	0.03
arXiv Megagauss	75p	36 [{36, 36}]	0.660 [0.522, 0.776]	32 [med 30, {28, 32}]	0.680 [0.542, 0.792]	+0.020†	9.46
arXiv Perovskite	80p	2	0.900 [0.786, 0.957]	3	0.740 [0.604, 0.841]	−0.160†	0.15

C Cross-vendor details

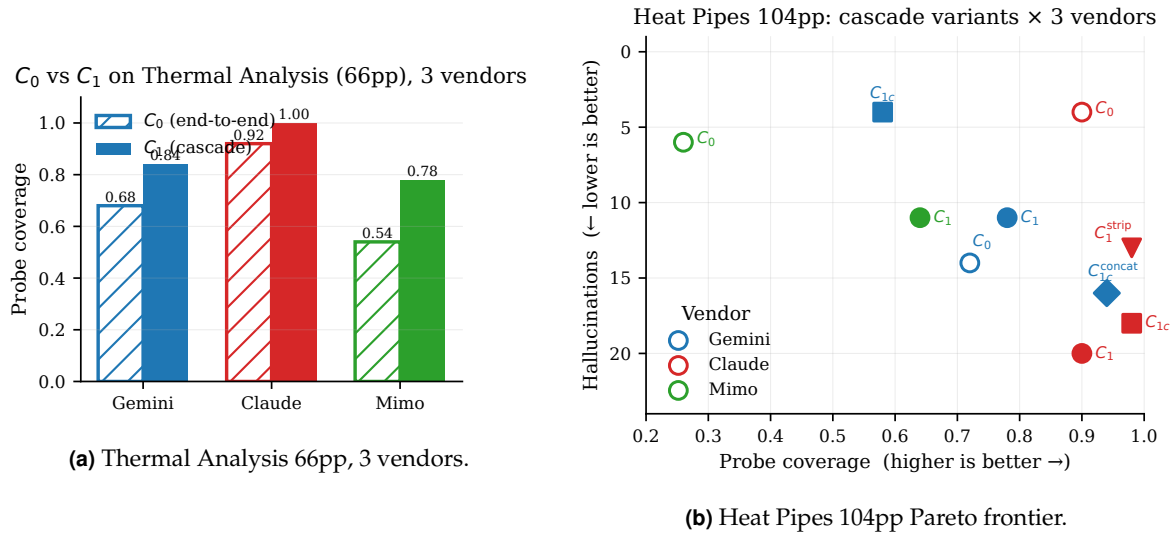


Figure 9. (a) On Thermal Analysis the cascade improves coverage on every vendor: Gemini $0.68 \rightarrow 0.84$, Claude $0.92 \rightarrow 1.00$, Mimo $0.54 \rightarrow 0.78$. (b) On Heat Pipes 104pp the three vendors give three patterns. Gemini cascade is small-Pareto-positive. Claude C_1 regresses on faithfulness (4h $\rightarrow$ 20h) leaving C_0 on the Pareto frontier. Mimo trades +5 halluc for +38pp coverage.

SP-5100 176pp replication. NASA SP-5100 “Shock and Vibration Technology” (1972, 176pp, 47k pdftotext words, 4.00 citation surface forms / 1k words) was chosen as a second long citation-heavy NASA survey for Mode B replication. Gemini C_0 median 46h/0.52 cov $\rightarrow$ C_1 median 45h/0.60 cov (range {25, 45, 59} across three judge re-runs of the byte-identical review, i.e. judge variance, not Pass-2 variance). Claude C_1 median 85h/0.81 cov (range {47, 85, 85, 97} across four judge re-runs, where the originally reported 47h was a low-end outlier). Mimo C_1 60h/0.23 cov. Claude C_0 not realizable: 176 page-images exceed Anthropic’s per-request payload limit even with the 1M-context beta. Spot-checking Claude C_1 shows the same bibliographic-expansion signature as Heat Pipes.

Modern arXiv reviews. Gemini Fairness AI: 14h/0.50 $\rightarrow$ 15h/0.40 (−10pp coverage, Mode A). Claude C_1 23h/0.70, and Mimo C_1 18h/0.32. Gemini Megagauss: C_0 median 36h/0.66 $\rightarrow$ C_1 median 30h/0.68 (high baseline-halluc physics review). Claude C_1 17h/0.92, and Mimo C_1 19h/0.26. Gemini Perovskite: 2h/0.90 $\rightarrow$ 3h/0.74 at $n=1$ (−6pp under multi-seed, near-noise-floor). Claude C_1 0h/0.90, and Mimo C_1 2h/0.90. The arXiv pattern (Mode A on 2/3) is opposite-direction from the NASA pattern despite comparable page counts. We attribute this to (a) arXiv full-color 200 DPI rendering carrying more visual signal per page, and (b) higher typesetting density forcing more aggressive C_1 compression at the same output budget.

D Cross-model audio: Gemini 2.5 Flash (clean, no bitrate confound)

Table 9 reports a full-quality C_0 -vs- C_1 comparison with Gemini 2.5 Flash as the subject model (same reused Pro reference and probes, same GPT-5.4 judge), over five representative audio

Table 8. Complete within-Claude C_0 -vs- C_1 comparison on the six NASA papers where Claude Opus 4.7’s multimodal C_0 fits the request payload. Halluc. is the GPT-5.4 unsupported-claim count; cov. is probe-coverage on the reused Gemini reference/probes. Cascade reduces hallucinations on 5/6 (all but the Heat Pipes Mode-B cell); coverage gain tracks C_0 headroom (flat where C_0 is already near-ceiling, a clean Pareto win on Thermal where C_0 has room). Cells sorted by length.

Cell	len	C_0 h/cov	C_1 h/cov	Δ_h	Δ_{cov}
Wagging Tail	9p	5 / 0.959	2 / 0.980	−3	+0.021
Env Test	18p	3 / 1.000	2 / 1.000	−1	±0.000
Dynamic Response	41p	2 / 1.000	1 / 0.960	−1	−0.040
Natural Vibration	42p	5 / 0.915	4 / 0.851	−1	−0.064
Thermal Analysis	66p	11 / 0.920	4 / 1.000	−7	+0.080
Heat Pipes	104p	4 / 0.900	20 / 0.900	+16	±0.000
<i>direction (noise floor $\Delta_h \geq 1, \Delta_{cov} \geq 0.04$)</i>				5/6 ↓	—

cells. Unlike the Mimo audio arm, the audio is the full 32 kbps source (no OpenRouter 10MB cap), so there is no audio-quality confound. 2.5 Flash is a strong end-to-end audio reader: its C_0 coverage exceeds Gemini 3.1 Pro’s on four of five cells, leaving little headroom. The cascade direction then follows the inverse-baseline law—a clear win on the two headroom cells (MIT, Harvard), flat on the saturated NeurIPS panel, and a faithfulness regression on Karpathy (where C_0 is already 0.92/3h). Net: faithfulness improves on 3/5 and coverage on 1/5, a smaller and headroom-gated effect rather than the uniform audio win seen on Pro. Gemini 3 Flash was attempted as a second subject but its verbatim Pass-1 transcription returns empty under the model’s recitation filter on copyrighted audio, so the cascade is not realizable on it.

Table 9. Gemini 2.5 Flash audio C_0 -vs- C_1 (five cells, full-quality audio, GPT-5.4 judge on the reused Pro reference/probes). Sorted by length. The win tracks C_0 headroom: large where C_0 is weak (MIT, Harvard), absent where C_0 is already strong (NeurIPS, Karpathy).

Cell	min	C_0 h/cov	C_1 h/cov	Δ_h	Δ_{cov}
3B1B Attention	26	5 / 0.900	1 / 0.920	−4	+0.020
NeurIPS panel	31	5 / 0.980	6 / 0.959	+1	−0.020
Karpathy	42	3 / 0.922	9 / 0.902	+6	−0.020
MIT 6.034	47	7 / 0.833	4 / 0.958	−3	+0.125
Harvard Moot	82	19 / 0.600	12 / 0.620	−7	+0.020

E Engineering interventions: chunked Pass-2 and citation stripping

Mode-B fix replication on 3 new long Claude cells. Table 11 reports the strip intervention across four Claude cells. On SP-5100, the published single-call snapshot $C_1 47h \rightarrow C_1^{\text{strip}} 24h$ initially looked like a clean halving, but the unstripped C_1 review judged four times yields {47, 85, 85, 97} (median 85), so the −23h effect compares one strip call against a low-end outlier of the unstripped distribution. On Megagauss, strip has essentially no effect (17 → 19h, 0.92 → 0.94 cov). On Fairness AI only 1 surface form matched our regexes, and the “stripped” Pass-2 produced a substantially different review (23 → 41h, 0.70 → 0.62 cov), revealing that

Table 10. Heat Pipes Claude 104pp variants. Strip dominates plain C_1 at $n=1$ on both axes. Chunked combinations do not improve over strip alone. At $n=5$ multi-seed, the strip effect on C_1 hallucinations is statistically null ($\hat{\Delta} = -1.8 \pm 2.6$, 95% CI $[-6.9, +3.3]$).

Variant	Halluc.	Coverage	Words
C_0 (image baseline)	4	0.900	3,308
C_1 (single Pass-2)	20	0.900	2,077
C_1^{strip} (citations $\rightarrow$ [REF])	13	0.980	2,471
C_{1c} (chunked + merge)	18	0.980	2,769
C_{1c}^{concat} (chunked, no merge)	20	0.940	2,777
C_{1c}^{strip} (strip + chunked + merge)	18	0.980	2,673
$C_{1c}^{\text{strip+concat}}$ (strip + chunked, no merge)	16	0.980	2,766

Claude’s text-only Pass-2 is itself non-deterministic. **Net Mode-B-fix status: zero replicated cells** (Heat Pipes retracted at $n=5$, Megagauss null, Fairness AI null-by-design, SP-5100 single-call-each).

Table 11. Mode B citation-strip intervention across four long Claude cells. “surface forms” = regex hits replaced with [REF].

Cell	len	surface forms	C_1 h	C_1 cov	C_1^{strip} h	C_1^{strip} cov
Heat Pipes (recall)	104p	55	20	0.900	13	0.980
SP-5100 (new)	176p	193	47	0.812	24	0.708
arXiv Megagauss (new)	75p	298	17	0.920	19	0.940
arXiv Fairness AI (new)	53p	1	23	0.700	41	0.620

Mode-A chunked Pass-2 replication and multi-seed verification. $n=1$ ($T=0$) results are consistent across four papers (Table 12): C_{1c}^{concat} (chunked, no merge) recovers coverage on long sources at a hallucination cost, while C_{1c} (chunked + merge) over-compresses: on Perovskite and Megagauss the merge variant collapses to 0.10 and 0.12 coverage respectively. At $n=5$ multi-seed (Table 13), the chunked-Pass-2 direction survives on Fairness AI and Perovskite but *not* on Megagauss, whose original $n=1$ (0.74cov) sits above the entire $n=5$ envelope $[0.64, 0.68]$ at $T=0.7$. We retract the original “chunked Pass-2 lifts coverage on all three arXiv counter-cells” claim.

Table 12. Chunked-Pass-2 ($n=1$, $T=0$) across four Gemini long-paper cells.

Cell	len	C_1 h/cov	C_{1c} h/cov	C_{1c}^{concat} h/cov	$\Delta_{\text{cov}}^{\text{concat}}$
Heat Pipes (recall)	104p	11 / 0.78	4 / 0.58	16 / 0.94	+0.16
arXiv Perovskite (new)	80p	3 / 0.74	1 / 0.10	13 / 0.82	+0.08
arXiv Megagauss (new)	75p	32 / 0.68	0 / 0.12	36 / 0.74	+0.06
arXiv Fairness AI (new)	53p	15 / 0.40	5 / 0.32	37 / 0.52	+0.12

Citation stripping on Gemini Natural Vibration ($n=5$ verified). At $n=1$ ($T=0$), strip moves halluc 5 $\rightarrow$ 12 but coverage 0.49 $\rightarrow$ 0.81 (+32pp). At $n=5$ ($T=0.7$, seeds 1–5): halluc

Table 13. Multi-seed verification of C_{1c}^{concat} on the three arXiv cells ($n=5$ seeds, Gemini 3.1 Pro Pass-2, $T=0.7$).

Cell	C_1 baseline	orig $n=1$	$n=5$ halluc	$n=5$ cov	cov range	dir. survives
arXiv Fairness AI	15h / 0.40	37h / 0.52	31.6 ± 1.9	0.496 ± 0.055	[0.440, 0.580]	yes
arXiv Megagauss	32h / 0.68	36h / 0.74	42.2 ± 4.3	0.664 ± 0.017	[0.640, 0.680]	NO
arXiv Perovskite	3h / 0.74	13h / 0.82	16.6 ± 2.5	0.832 ± 0.033	[0.800, 0.880]	yes

{15, 11, 9, 8, 8} with mean 10.2, and coverage {0.660, 0.723, 0.702, 0.511, 0.766} with mean 0.672. The strip-vs- C_1 direction (cov gain at halluc cost) survives all 5 seeds. The magnitude on coverage is +18pp (mean) versus the original +32pp single-seed reading.

F Multi-seed envelopes on the four headline cells

Table 14. Per-cell multi-seed envelopes for the four headline cells (Gemini C_1 Pass-2 at $T=0.7$, seeds 1–5). “base-in-envelope?” checks whether the $n=1$ value sits within the per-seed range.

Cell	orig $n=1$	$n=5$ halluc	$n=5$ cov	base-in-envelope?
MIT 6.034 Winston	4h / 0.98	7.4 ± 1.5	0.971 ± 0.028	cov yes; halluc NO (low)
Sapolsky Behavioral	15h / 0.94	14.5 ± 2.5	0.898 ± 0.017	halluc yes; cov NO (high)
Heat Pipes Gemini	11h / 0.78	15.4 ± 2.7	0.732 ± 0.056	cov yes; halluc NO (low)
arXiv Perovskite	3h / 0.74	4.8 ± 2.2	0.844 ± 0.056	halluc yes; cov NO (low)

Two readings: (1) the published $T=0$ baselines are mostly representative of the $T=0.7$ distribution on the headline cells, since every cell’s $n=5$ envelope overlaps the published number on at least one axis. (2) Perovskite was the single seed-envelope point most far from the multi-seed mean (-0.10 pp coverage below it), and under multi-seed verification its cascade-cov loss reads -6 pp rather than -16 pp. We retain Perovskite as a near-noise-floor cell and treat its earlier “Mode-A counter-cell” label as overstated at $n=1$.

G E6 vision-model ladder and parametric load sweep

Table 15 reports the E6 logit-lens readout for both open VLMs across the load sweep $K \in \{1, 4, 8, 16, 24\}$ (facts packed into one stimulus, one queried). Answer accuracy is 1.00 for text and image at every K on both models, and the text–image readout-depth gap stays within ± 1 layer throughout (TOST-equivalent within a ± 1 -layer SESOI at every row). The Qwen2.5-VL-32B variant was excluded: it verbalizes the answer in a form our single-token matcher cannot capture (a spurious acc = 0 artifact), so its readout depths are unreliable, and the 3B and 7B models carry the claim.

H Three sources of stochasticity

Detailed decomposition. (a) *Claude Pass-2 generation variance.* Same input + same API + same prompt: Heat Pipes Claude C_1 swings $20 \rightarrow 9$ between two runs (strip $13 \rightarrow 14$ more stable), and the deltas are large enough that the strip effect cannot be distinguished from Pass-2 noise. Claude Opus 4.7 deprecates the temperature parameter so determinism cannot

Table 15. E6 cross-modal logit-lens across two open VLMs and a K -fact load sweep. *text/img depth* = first layer the answer token is top-1. *gap CI90* = 90% CI on (img–text) depth in layers. *equiv* = TOST-equivalent within ± 1 layer. Accuracy is 1.00 for both modalities in every row.

Model	K	text acc	img acc	text depth	img depth	gap CI90 (equiv)
Qwen2.5-VL-3B	1	1.00	1.00	32.00	32.00	[0.00, 0.00] (yes)
Qwen2.5-VL-3B	4	1.00	1.00	32.25	32.06	[−0.41, 0.04] (yes)
Qwen2.5-VL-3B	8	1.00	1.00	32.19	32.06	[−0.27, 0.02] (yes)
Qwen2.5-VL-3B	16	1.00	1.00	32.75	32.06	[−0.88, −0.49] (yes)
Qwen2.5-VL-3B	24	1.00	1.00	32.62	32.00	[−0.83, −0.42] (yes)
Qwen2.5-VL-7B	1	1.00	1.00	24.31	24.44	[−0.02, 0.27] (yes)
Qwen2.5-VL-7B	4	1.00	1.00	24.25	24.12	[−0.27, 0.02] (yes)
Qwen2.5-VL-7B	8	1.00	1.00	24.00	24.00	[−0.15, 0.15] (yes)
Qwen2.5-VL-7B	16	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)
Qwen2.5-VL-7B	24	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)

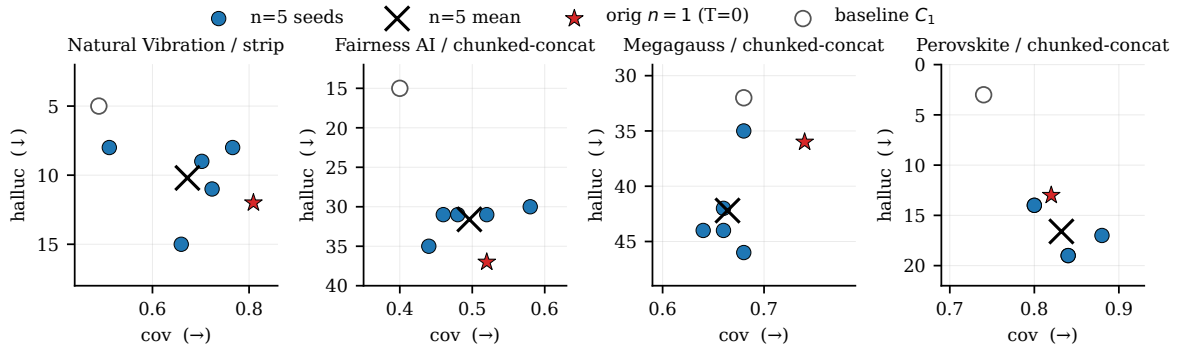


Figure 10. $n=5$ multi-seed envelopes for four intervention-verification cells. Blue dots: five Pass-2 seeds at $T=0.7$. Black $\times$: $n=5$ mean. Red star: originally-published $n=1$ at $T=0$. Open circle: cell’s C_1 baseline (no intervention). NV strip, Fairness AI, Perovskite: $n=1$ sits within $n=5$ envelope. Megagauss: $n=1$ is outside on both axes (lower halluc and higher cov than any of 5 seeds), so the published Megagauss chunked-Pass-2 cov-gain claim is retracted at $n=5$.

be enforced from the SDK. (b) *Gemini Pass-2 generation variance*. At temperature=0.0, two of three Gemini Pass-2 re-runs produced byte-identical output (MD5 match) on Heat Pipes / SP-5100, and the third differed in similar length (1929 vs 1815 words, $\Delta h = +6$). Gemini is therefore mostly but not strictly deterministic, consistent with GPU-level non-associativity. (c) *GPT-5.4 high-reasoning judge variance*. Essentially deterministic on low-count cells (± 1 on Heat Pipes Claude across three re-judgments) but substantially stochastic on high-count cells (± 25 on SP-5100, ± 17 on the byte-identical SP-5100 Gemini review). We attribute this to long unsupported-claim lists running up against our 24k-token completion budget, so the judge truncates and resamples across calls. Coverage is stable on every cell across judge re-runs.

$n=5$ multi-seed verification on Heat Pipes Claude. C_1 halluc {9, 11, 11, 15, 20} (mean 13.2, stdev 4.4, range 9–20), and C_1^{strip} halluc {5, 11, 13, 14, 14} (mean 11.4, stdev 3.8, range 5–14). Coverage means identical (0.936 both) with similar spread. Strip effect on hallucinations at $n=5$: $\hat{\Delta} = -1.8$, $\text{SE} \approx 2.6$, 95% CI [−6.9, +3.3], which crosses zero. **The Mode B fix is**

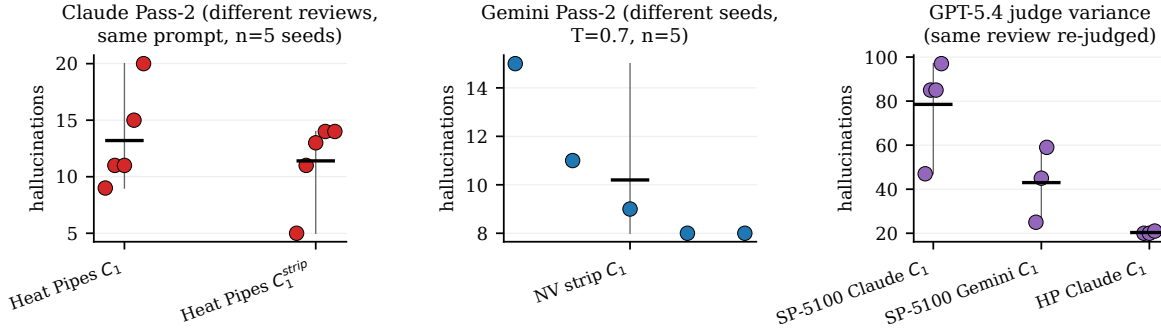


Figure 11. Three sources of stochasticity, decomposed. *Left:* Claude Pass-2 generation variance, with 5 reruns on Heat Pipes yielding {9, 11, 11, 15, 20} (range 11). *Middle:* Gemini Pass-2 generation variance at $T=0.7$ on Natural Vibration strip: {15, 11, 9, 8, 8} (range 7). *Right:* GPT-5.4 judge variance on the *same* review file judged multiple times. SP-5100 Claude C₁ produces {47, 85, 85, 97} over 4 judge re-runs (range 50), while Heat Pipes Claude C₁ is stable across re-judgments. Coverage is much more stable than hallucination counts across all three noise sources.

statistically null on Claude/Heat Pipes at $n=5$.

Table 16. Stochasticity decomposition. Top: Claude Pass-2 generation variance (different reviews, same prompt, judged once each). Bottom: GPT-5.4 judge variance on the *same* Claude C₁ review file (one review, judged multiple times).

Cell	Condition	Run 1	Run 2	Δh	Δcov
<i>Claude Pass-2 generation variance</i>					
Heat Pipes 104p	C ₁	20h / 0.90	9h / 0.90	-11	± 0.00
	C ₁ ^{strip}	13h / 0.98	14h / 0.88	+1	-0.10
SP-5100 176p	C ₁	47h / 0.81	79h / 0.81	+32	± 0.00
	C ₁ ^{strip}	24h / 0.71	97h / 0.77	+73	+0.06
Fairness AI 53p	C ₁	23h / 0.70	27h / 0.66	+4	-0.04
	C ₁ ^{strip}	41h / 0.62	17h / 0.60	-24	-0.02
<i>GPT-5.4 judge variance (same review, judged multiple times)</i>					
Heat Pipes 104p	C ₁ (calls 1-3)	20h / 0.90	21h / 0.92	+1	+0.02
SP-5100 176p	C ₁ (calls 1-4)	47h / 0.81	97h / 0.81	+50	± 0.00

I Inter-judge details

Per-probe agreement (the fraction of probes the two judges classify identically as COVERED / MISSING) ranges 0.882–1.000 across the (cell, condition) pairs, median ≈ 0.94 . Two of the three hallucination-direction disagreements on the original 10 cells (Wagging Tail, Env Test) are near-tie cells with ≤ 5 hallucinations under both judges. The third (SNAC) is driven by the judges assigning different C₀ baselines (14 vs 10) while agreeing on C₁.

J Probe-circularity and reference-bias check tables

Table 17. Inter-judge hallucination counts on the original 10 cells. Direction $\checkmark$ means both judges agree on the sign of $C_0 \rightarrow C_1$. (Per-cell deltas for the new 11 cells are saved in `research_log/inter_judge_summary_new12.json`.)

Cell	GPT-5.4		Claude		Direction agree?
	C_0	C_1	C_0	C_1	
Karpathy review	13	3	7	4	$\checkmark$
Karpathy mtg-min	4	2	1	0	$\checkmark$
SNAC mtg-min	14	11	10	11	$\times$
Wagging Tail (9pp)	3	1	2	2	$\times$
Env Test (18pp)	5	4	0	2	$\times$
Dynamic Response (41pp)	3	2	1	0	$\checkmark$
Natural Vibration (42pp)	4	5	2	4	$\checkmark$
Thermal Analysis (66pp)	5	2	2	1	$\checkmark$
Heat Pipes (104pp)	14	11	12	11	$\checkmark$
Thermal Analysis Claude	11	4	9	4	$\checkmark$

Table 18. Inter-judge probe-coverage counts (probes covered out of n) on the original 10 cells.

Cell	n	GPT-5.4		Claude		Direction agree?
		C_0	C_1	C_0	C_1	
Karpathy review	51	32	49	38	49	$\checkmark$
Karpathy mtg-min	50	8	17	13	21	$\checkmark$
SNAC mtg-min	49	14	26	15	28	$\checkmark$
Wagging Tail (9pp)	49	39	38	40	38	$\checkmark$
Env Test (18pp)	50	44	46	46	49	$\checkmark$
Dynamic Response (41pp)	50	29	35	32	36	$\checkmark$
Natural Vibration (42pp)	47	31	23	32	28	$\checkmark$
Thermal Analysis (66pp)	50	34	42	34	45	$\checkmark$
Heat Pipes (104pp)	50	36	39	39	41	$\checkmark$
Thermal Analysis Claude	50	46	50	46	49	$\checkmark$

Table 19. Probe-circularity check on 6 cells. “Original” = published numbers (Gemini-extracted probes, GPT-5.4 judge).

Cell (orig $\rightarrow$ Claude-extracted probes)	Δ_{cov} (orig)	Claude-pr. / GPT	Claude-pr. / Claude
Karpathy review (51 $\rightarrow$ 55) [audio]	+0.333	+0.400	+0.236
MIT 6.034 (48 $\rightarrow$ 50) [audio]	+0.208	+0.180	+0.120
Heat Pipes 104pp (50 $\rightarrow$ 60) [paper]	+0.060	+0.050	+0.033
Natural Vibration 42pp (47 $\rightarrow$ 50) [paper]	-0.170	-0.160	-0.200
arXiv Fairness AI 53pp (50 $\rightarrow$ 50) [paper]	-0.100	-0.100	+0.040
arXiv Megagauss 75pp (50 $\rightarrow$ 60) [paper]	+0.020	+0.000	+0.000
arXiv Perovskite 80pp (50 $\rightarrow$ 54) [paper]	-0.160	-0.167	-0.111

K Cascade cost

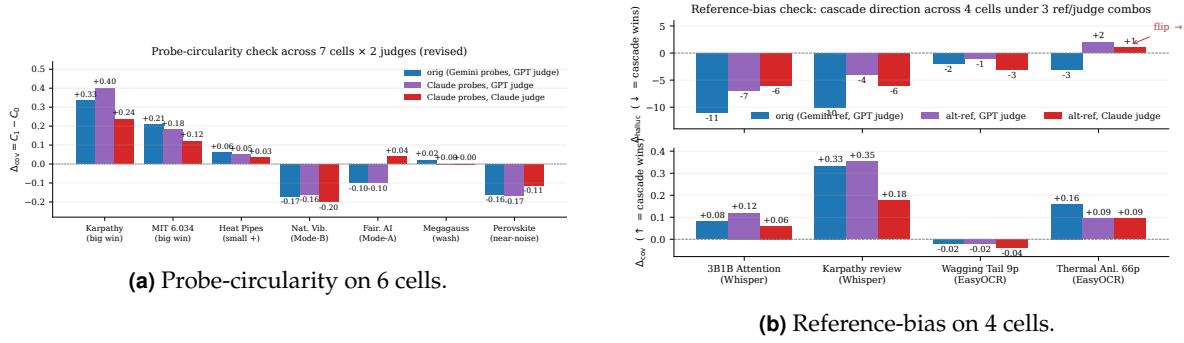


Figure 12. (a) Cascade Δ_{cov} under three (probe-extractor, judge) combinations. Sign is preserved on 6/6 under GPT-5.4 and 5/6 under Claude (Fairness AI flips, both within noise). (b) Cascade direction with non-Gemini reference (Whisper-base on 3B1B + Karpathy, EasyOCR on Wagging Tail + Thermal Analysis). Coverage direction preserved on 4/4 under both judges. Hallucination direction flips on Thermal Analysis under EasyOCR, a reference-side artifact (EasyOCR loses figure captions).

Table 20. Reference-bias check on 4 cells. The non-Gemini reference rows replace OCR/ASR only, and probes are re-extracted by Claude. “orig” = Gemini reference + GPT-5.4 probes + GPT-5.4 judge.

Cell	Reference	Δ_h orig	Δ_h alt	Δ_{cov} orig	Δ_{cov} alt
3B1B Attention 26m	Whisper-base / GPT-5.4	-11	-7	+0.080	+0.120
3B1B Attention 26m	Whisper-base / Claude	-11	-6	+0.080	+0.060
Karpathy review 42m	Whisper-base / GPT-5.4	-10	-4	+0.333	+0.353
Karpathy review 42m	Whisper-base / Claude	-10	-6	+0.333	+0.176
Wagging Tail 9p	EasyOCR / GPT-5.4	-2	-1	-0.020	-0.019
Wagging Tail 9p	EasyOCR / Claude	-2	-3	-0.020	-0.038
Thermal Analysis 66p	EasyOCR / GPT-5.4	-3	+2	+0.160	+0.094
Thermal Analysis 66p	EasyOCR / Claude	-3	+1	+0.160	+0.094

Table 21. Hallucination density (per 1k output words) on Thermal Analysis 66pp.

	C_0	C_1
Gemini 3.1 Pro	4.2	1.4
Claude Opus 4.7	3.7	1.4
Mimo v2-omni	4.9	5.4

L Per-cell hallucinations per 1k output words

End-to-end hallucination rates differ per vendor (3.7 / 4.2 / 4.9 per 1k), and cascade brings Gemini and Claude to a common ≈ 1.4 per 1k floor on the moderate-length paper. Mimo does not converge to this floor.

Claude’s per-1k hallucination rate jumps from 1.2 (C_0) to 9.6 (C_1) on Heat Pipes, an $8\times$ increase that survives length normalization. The same model on the same protocol on Thermal 66pp moved $3.7 \rightarrow 1.4$. Citation stripping pulls Heat Pipes Claude back to 5.3/1k. Gemini and Mimo on the same paper sit at 6–8/1k regardless of cascade variant, indicating that the 1.4/1k floor reached on Thermal is not a property of the cascade in general, but of the specific (model, source) interaction.

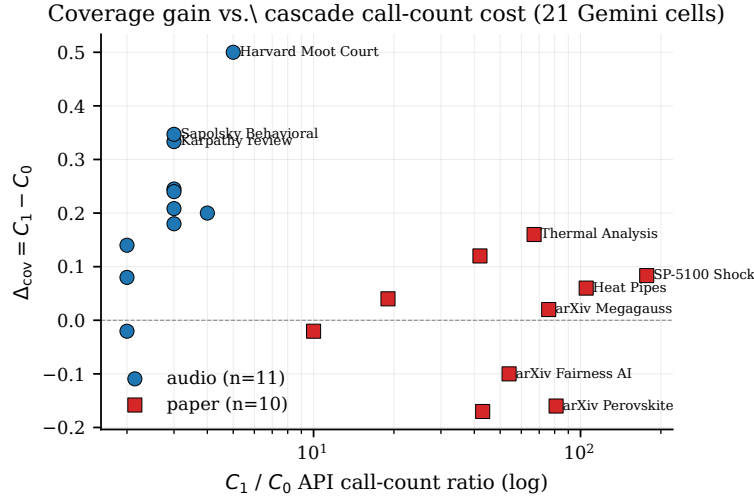


Figure 13. Cascade coverage gain vs. API call-count cost. Each point is one Gemini cell. The C_1/C_0 call ratio for paper cells is $N + 1$ (page-chunked OCR + one Pass-2) and ranges from $10\times$ for a 9-page paper to $177\times$ for SP-5100. Audio cells use 30-min chunked ASR (ratio $\approx 2\text{--}3\times$). Cascade buys most of its coverage gain at the audio end and small or negative gains at the high-cost paper end. For example, Perovskite at $81\times$ call ratio loses 16 pp of coverage (revised to -6pp under multi-seed). SP-5100 at $177\times$ has C_0 baseline so low (0.52) that the cascade still wins on coverage ($+8\text{pp}$) but at the highest cost in the dataset.

Table 22. Hallucination density (per 1k output words) on Heat Pipes 104pp. Claude C_1 at 9.6/1k is the largest value in our dataset and dwarfs Claude’s Thermal C_1 rate of 1.4/1k, supporting the Mode B interpretation as a length- and citation-density-triggered effect rather than a per-token verbosity effect.

	C_0	C_1
Gemini 3.1 Pro	7.6	6.1
Claude Opus 4.7	1.2	9.6
Claude Opus 4.7 (C_1^{strip})	—	5.3
Mimo v2-omni	7.0	6.9

M Specific hallucination examples (Heat Pipes Claude)

The 20 unsupported claims from Claude C_1 on Heat Pipes 104pp are dominated by training-prior expansions of bibliographic abbreviations:

- “Cotter’s 1965 analysis (LA-3246-MS)”: the transcript cites only “Reference 2 (Cotter)”, so the year and report number are training-data details.
- “correlation from Allingham and McEntire (1961)”: the transcript only says “Ref. 12”, so the author names and year are training-data details.
- “McAdams’ laminar horizontal-tube formula”: the transcript says “Reference 13 for laminar condensation on horizontal tubes”, so “McAdams” is a training-data detail.
- “Vapor chamber-fin radiator (Haller, Lieblein, and Lindow at NASA Lewis)”: the transcript says only “proposed by Haller, et al of NASA”.

One outright misreading: the transcript states the no-boiling condition as $p_v - p_\ell \leq$

$2\sigma \cos \theta / r_c$, but Claude’s review writes $\geq$. This is a sign error in Pass-2 reading of Pass-1 text, not training-prior leak.

N Code and data availability

All code, prompts, per-cell judgments, and the mechanistic-experiment artifacts (including the E8 substrate probe and the within-Claude cross-vendor comparison) are released in the project repository, whose README documents the pipelines and the exact commands to reproduce every cell, table, and figure in this paper.